

A LAZARD-TYPE CORRESPONDENCE AND APPLICATIONS TO SKEW BRACES AND POST-LIE RINGS

S. TRAPPENIERS

ABSTRACT. Through a Hopf algebraic approach, we develop a framework for formal differentiation and integration and prove a correspondence theorem that comprises the Lazard correspondence, the Malcev correspondence and the formal integration of complete connected Lie algebras with characteristic 0. We then apply this theory to obtain a correspondence theorem for classes of left nilpotent skew braces on a nilpotent group and left nilpotent post-Lie rings on a nilpotent Lie ring.

1. INTRODUCTION

In 1954, Lazard proved a correspondence between Lie rings and groups satisfying certain completeness conditions [21]. The nowadays better known version of this, although more restrictive than the original version, states that for p^k a prime power, there is a bijective correspondence between nilpotent Lie rings of order p^k and nilpotency class at most $p - 1$, and nilpotent groups of order p^k of nilpotency class at most $p - 1$. In particular, one finds a correspondence between groups of order p^k and nilpotent Lie rings of order p^k for $k < p$. This has proved to be very useful in the classification of finite p -groups, see for example [25]. Lazard's ideas built upon earlier work by Malcev [22], who proved a correspondence between nilpotent rational Lie algebras and nilpotent radicable torsionfree groups. Quillen extended Malcev's result to arbitrary field of characteristic 0 [28], his technique is now generally referred to as formal integration.

Pre-Lie algebras, also called left-symmetric algebras, arose as early as in 1881 in the work of Cayley [11], only to resurface again 80 years later in the works of Vinberg [40] and Koszul [20]. See [9] for a survey on pre-Lie algebras. One of the motivations to study pre-Lie algebras comes from geometry, as they appear naturally in the study of regular affine actions of Lie groups [19]. More recently, the notion of a post-Lie algebra was introduced in the works of Valette [39], see also [15]. For R a commutative ring, a *post-Lie R -algebra* is a Lie R -algebra $\mathfrak{a}$ together with an R -bilinear map $\triangleright : \mathfrak{a} \times \mathfrak{a} \rightarrow \mathfrak{a}$ satisfying:

$$(1.1) \quad x \triangleright [y, z] = [x \triangleright y, z] + [y, x \triangleright z],$$

$$(1.2) \quad [x, y] \triangleright z = (x, y, z)_{\triangleright} - (y, x, z)_{\triangleright},$$

with $(x, y, z)_{\triangleright} := x \triangleright (y \triangleright z) - (x \triangleright y) \triangleright z$. If $\mathfrak{a}$ is abelian, then $(\mathfrak{a}, \triangleright)$ is a *pre-Lie R -algebra*. A post-Lie $\mathbb{Z}$ -algebra, respectively pre-Lie $\mathbb{Z}$ -algebra is called a *post-Lie ring*, respectively *pre-Lie ring*.

Braces were introduced in 2006 by Rump [29, 30] in the study of the set-theoretical Yang–Baxter equation. They were subsequently generalized by Guarnieri and Vendramin to skew braces [18]. A skew brace is a triple $(A, \cdot, \circ)$ with A a set

2020 *Mathematics Subject Classification.* 17B69, 20N99, 20F19, 20F40.

Key words and phrases. post-Lie algebra, pre-Lie algebra, brace, skew brace, left nilpotent, nilpotent group, Lazard correspondence.

and $(A, \cdot)$ and $(A, \circ)$ group structures such that

$$a \circ (b \cdot c) = (a \circ b) \cdot a^{-1} \cdot (a \circ b),$$

holds for all $a, b, c \in A$. Here a^{-1} denotes the inverse of a in $(A, \cdot)$. When $(A, \cdot)$ is abelian, then $(A, \cdot, \circ)$ is called a *brace*. For braces, the notation $(A, +, \circ)$ is also common (in fact, this is also the case for skew braces, but to avoid any possible confusion we will abstain from using $+$ for a non-abelian group operation). Skew braces are essentially the same as bijective 1-cocycles of groups or regular subgroups of the holomorph of a group. In this sense, they have already been studied in a geometric framework in the form of regular affine actions of Lie groups on vector spaces [3, 24] long before their current definition. Since pre-Lie algebras played an important role there, some connection between skew braces and pre-Lie rings is to be expected. This was indeed first stated by Rump [31] for $\mathbb{R}$ -braces, he also gave an idea of how to use the Lazard correspondence for the finite case. Bachiller then used techniques similar to the ones that Rump alluded to in order to give a counterexample of a conjecture on braces [4], starting from a counterexample by Burde [8] to a question by Milnor [24] (note that the first counterexample to Milnor's question was given earlier by Benoist [6]).

Smoktunowicz proved that the construction of the group of flows, as defined for pre-Lie algebras in [2], could be adapted to left nilpotent pre-Lie rings of prime power order p^k , with $k+1 < p$, in order to obtain braces [36] with the same additive structure. Here, a post-Lie algebra $(\mathfrak{a}, \triangleright)$ is *left nilpotent* if there exists some n such that $\mathfrak{a}^n = 0$ where $\mathfrak{a}^1 = \mathfrak{a}$ and $\mathfrak{a}^{i+1}$ is the subgroup of $(\mathfrak{a}, +)$ generated by $a \triangleright b$ for $a \in \mathfrak{a}$, $b \in \mathfrak{a}^i$, *right nilpotency* is defined analogously. Similarly, a skew brace $(A, \cdot, \circ)$ is *left nilpotent* if there exists some n such that $A^n = 1$, where $A^1 = A$ and A^{i+1} is the subgroup of $(A, \cdot)$ generated by the elements $a * b = a^{-1} \cdot (a \circ b) \cdot b^{-1}$ for all $a \in A$, $b \in A^i$, *right nilpotency* is defined analogously. Recall that a skew brace of prime power order is always left nilpotent [12]. It is not immediately clear whether this construction is invertible, as only an inverse construction is given in the case of strongly nilpotent braces of order p^k , with $k+1 < p$, and with strong nilpotency index at most $p-1$. Here strong nilpotency is equivalent to the structures being both left and right nilpotent [33]. This correspondence was used to classify all strongly nilpotent braces of order p^4 , for $p > 5^5$, in [26] while the non-strongly nilpotent ones were classified earlier in [27]. In [35], a construction is given starting from a brace $(A, +, \circ)$ of order p^k , with $k+1 < p$, in order to obtain a pre-Lie ring structure with additive group $(A, +)/\{a \in A \mid p^2a = 0\}$. Although the strong nilpotency condition is no longer present, one can not hope that this yields a correspondence as we quotient by a non-trivial subgroup.

Recently, Bai, Guo, Sheng and Tang showed how from a regular affine action of a Lie group onto another Lie group, one obtains a post-Lie algebra through differentiation [5] (note that, although not explicitly stated in this way, this construction also appears in [10]). This extended the earlier techniques to obtain a pre-Lie algebra in the case that the latter group is $\mathbb{R}^n$. Conversely, also in [5] a construction by formal integration was given which yields a skew brace structure starting from a post-Lie algebra of characteristic 0 and with some completeness conditions.

A common idea is present in the earlier mentioned works of Malcev, Lazard and Quillen: starting from a Lie ring $\mathfrak{g}$ we want to embed it into an algebra A , moreover the exponential and logarithmic map $\exp$ and $\log$ should be defined on (subsets of) A and in this way $\exp(\mathfrak{g})$ is a group (with some additional structure). This group structure is then transferred onto $\mathfrak{a}$ since $\log(\exp(\mathfrak{a})) = \mathfrak{a}$. The first goal of this paper is to introduce a general theory where precisely this idea can be applied. The second goal is then to define post-Lie algebras and skew braces in this context and give a uniform treatment of formal differentiation and integration, yielding a

correspondence between these two structures. The exact definitions can be found in the respective sections, but the following overview should give a general intuition as to how we approach this. In particular, note that the analogy with the approach taken in [28, 41].

In Section 2 we introduce the main actors of this paper, i.e. sequences of rings $\mathcal{R}$ and modules over such sequences. We prove the necessary properties of this category of modules which allow us to define $\mathcal{R}$ -algebras, Lie $\mathcal{R}$ -algebras, co $\mathcal{R}$ -algebras and Hopf $\mathcal{R}$ -algebras.

We next proceed in Section 3 to prove that if $\mathcal{R}$ has a certain divisibility property, then the Poincaré-Birkhoff-Witt theorem holds in our context.

By introducing a completion operation in Section 4 we can then also obtain the Poincaré-Birkhoff-Witt theorem in the complete setting and moreover also the Milnor-Moore theorem is proved under the earlier divisibility condition on $\mathcal{R}$.

In Section 5, we then introduce the notion of an $\mathcal{R}$ -group. They are naturally obtained as the grouplike elements of a (complete) Hopf $\mathcal{R}$ -algebra. By defining the exponential and logarithmic map on complete Hopf $\mathcal{R}$ -algebras we are able to obtain a Lazard-type correspondence between complete Lie $\mathcal{R}$ -algebras and complete $\mathcal{R}$ -groups under some natural conditions.

In Section 6 and Section 7 we introduce the notion of post-Lie $\mathcal{R}$ -algebras and skew $\mathcal{R}$ -braces. In particular, we prove if $(A, \cdot, \circ)$ is a left nilpotent skew brace and $(A, \cdot)$ is a nilpotent group, then also $(A, \circ)$ is a nilpotent group. This result was previously only known for finite skew braces [12, Theorem 4.8]. In fact we have a stronger result, for which we first have to introduce a new notion of nilpotency.

Let $(A, \cdot, \circ)$ be a skew brace, define inductively $L^1(A) = A$ and $L^{i+1}(A)$ the subgroup of $(A, \cdot)$ generated by all elements of the form $a * b = a^{-1} \cdot (a \circ b) \cdot b^{-1}$ and $a \cdot b \cdot a^{-1} \cdot b^{-1}$ for $a \in A$, $b \in L^i(A)$. Note $L^i(A)$ is the left hand version of the series $R_n(A, A)$ as defined in [1]. By induction, one sees that the $L^i(A)$ are a descending chain of normal subgroups of $(A, \cdot)$. We now define L -nilpotency of skew braces, which for braces clearly coincides with left nilpotency.

Definition 1.1. A skew brace $(A, \cdot, \circ)$ is *L-nilpotent* if there $L^{k+1}(A) = 1$ for some $k \geq 0$. The minimal such k , if it exists, is the *L-nilpotency class* of A .

We can now give a more precise formulation of our results which imply the earlier claim.

Lemma 1.2. *Let $(A, \cdot, \circ)$ be a skew brace. Then A is L-nilpotent if and only if A is left nilpotent and $(A, \cdot)$ is a nilpotent group.*

Theorem 1.3. *Let $(A, \cdot, \circ)$ be an L-nilpotent skew brace of class k . Then $(A, \circ)$ is nilpotent of class at most k and the k -th term in the lower central series of $(A, \circ)$ is contained in $\text{Ann}(A)$.*

Recall that

$$\begin{aligned} \text{Soc}(A) &:= \{a \in A \mid a \circ b = a + b = b + a \text{ for all } b \in A\}, \\ \text{Fix}(A) &:= \{a \in A \mid b \circ a = b + a \text{ for all } b \in A\}, \\ \text{Ann}(A) &:= \text{Soc}(A) \cap \text{Fix}(A). \end{aligned}$$

In Section 8 we first prove that the exp and log give a correspondence between certain derivations and automorphisms of Lie $\mathcal{R}$ -algebras, mimicking the geometric correspondence given by integration and differentiation. We then prove the correspondence between post-Lie $\mathcal{R}$ -algebras and skew $\mathcal{R}$ -braces under the necessary completeness and divisibility condition. In particular, we obtain an extension of Smoktunowicz's correspondence on pre-Lie rings and braces of prime power order. Before formulating this result, we introduce the notion of L -nilpotency.

Let $(\mathfrak{a}, \triangleright)$ be a post-Lie ring. Inductively define $L^1(\mathfrak{a}) = \mathfrak{a}$ and $L^{i+1}(\mathfrak{a})$ the subgroup of $(\mathfrak{a}, +)$ generated by all elements of the form $a \triangleright b$ and $[a, b]$ where $a \in A$, $b \in L^i(A)$. One easily verifies that $L^i(\mathfrak{a})$ gives a descending filtration of ideals of the Lie ring $\mathfrak{a}$. We now are ready to define L -nilpotency, which for pre-Lie rings is the same as left nilpotency.

Definition 1.4. A post-Lie ring $(\mathfrak{a}, \triangleright)$ is L -nilpotent if $L^{k+1}(\mathfrak{a}) = 0$ for some $k \geq 0$. The minimal such k , if it exists, is the L -nilpotency class of $(\mathfrak{a}, \triangleright)$.

We are now ready to formulate the result, which will follow as a somewhat direct application of the more technical correspondence in Theorem 8.19.

Theorem 1.5. *Let p be a prime and $k < p$. There is a bijective correspondence between post-Lie rings $(\mathfrak{a}, \triangleright)$ of L -nilpotency class k such that $(\mathfrak{a}, +)$ is a p -group and skew braces $(A, \cdot, \circ)$ of L -nilpotency class k such that $(A, \cdot)$ is a p -group.*

In particular, there is a bijective correspondence between skew braces of size p^k and L -nilpotent post-Lie rings of size p^k .

This correspondence is explicitly described in Proposition 8.14 and Proposition 8.17. The proof of Theorem 1.5 is given at the end of Section 8.

We further discuss how our notion of completeness relates to the classical notion of completeness for finite dimensional pre-Lie algebras defined in [19] and to the notion of completeness as coined in [5] and how our constructions related to the ones in [2, 5, 36].

In [5] the authors propose to find a suitable differentiation and formal integration theory for finite skew braces and post-Lie rings. On the one hand, the idea of considering the logarithm of function as a differentiation, which is the principal approach in this paper, does prove to be fruitful. On the other hand, some techniques used by Smoktunowicz in [34, 36] could also be seen as a discrete version of differentiation. Therefore, in Section 9 we motivate why this is the case and we give an application to skew braces that extends an application by Smoktunowicz. We end by emphasizing a possible advantage of this technique and ask whether this could possibly lead to a relaxation on the nilpotency condition imposed by the exponent of the group in [36, Theorem 13]. In particular, this could lead to applications to braces of size 2^k , whereas up until now all of the known constructions are trivial when $p = 2$.

2. THE CATEGORY OF $\mathcal{R}$ -MODULES

In what follows, $\mathcal{R}$ will always denote a sequence of commutative rings

$$R_0 \xrightarrow{d_0} R_1 \xrightarrow{d_1} R_2 \xrightarrow{d_2} R_3 \xrightarrow{d_3} \dots$$

where $d_i : R_i \rightarrow R_{i+1}$ are ring homomorphisms. All rings are assumed to be unital, but we do not exclude the case $0 = 1$. For a ring R we denote by $\overline{R}$ the constant sequence where $R_i = R$ and $d_i = \text{id}_R$ for all $i \geq 0$.

Definition 2.1. An $\mathcal{R}$ -module is an abelian group M with a descending filtration of subgroups $M_0 \supseteq M_1 \supseteq \dots$ such that for every $i \geq 0$, M_i is a R_i -module such that moreover $r \cdot_i m = d_i(r) \cdot_{i+1} m$ for all $r \in R_i$ and $m \in M_{i+1}$.

An $\mathcal{R}$ -submodule N of an $\mathcal{R}$ -module M is a subgroup of M with filtration $N_i = N \cap M_i$ such that N_i is a R_i -submodule of M_i for all $i \geq 0$. Given a submodule N we define the quotient module M/N as the quotient M/N (as an abelian group) with filtration $(M/N)_i = (M_i + N)/N$ and the natural $\mathcal{R}$ -module structure coming from the isomorphism $(M_i + N)/N \cong M_i/(N \cap M_i)$.

A *homomorphism* of $\mathcal{R}$ -modules is a map $f : M \rightarrow N$ such that for all $i \geq 0$, $f(M_i) \subseteq N_i$ and the restrictions $f : M_i \rightarrow N_i$ are R_i -module homomorphisms. The ring of all $\mathcal{R}$ -module homomorphisms of M is denoted $\text{End}_{\mathcal{R}}(M)$.

The category of $\mathcal{R}$ -modules is denoted ${}_{\mathcal{R}}\mathbf{Mod}$.

Note that because of the compatibility condition, we can denote all module actions by juxtaposition without any possible confusion. Another way to see an $\mathcal{R}$ -module is as follows: Let M be an abelian group with a descending filtration. For $i \geq 0$, define $\text{End}_f(M_i)$ as all group endomorphisms of M_i which map M_j into itself for $j \geq i$, hence we get restriction maps $\text{End}_f(M_i) \rightarrow \text{End}_f(M_{i+1})$. An $\mathcal{R}$ -module is then a family of ring homomorphisms $(\rho_i : R_i \rightarrow \text{End}_f(M_i))_{i \geq 0}$ such that the following diagram commutes.

$$\begin{array}{ccccccc} R_0 & \xrightarrow{d_0} & R_1 & \xrightarrow{d_1} & R_2 & \xrightarrow{d_2} & \dots \\ \downarrow \rho_0 & & \downarrow \rho_1 & & \downarrow \rho_2 & & \\ \text{End}(M_0) & \longrightarrow & \text{End}(M_1) & \longrightarrow & \text{End}(M_2) & \longrightarrow & \dots \end{array}$$

Example 2.2. For R a ring, an $\overline{R}$ -module is simply a filtered R -module.

Example 2.3. Given an R_i module M , for some $i \geq 0$, we can make M into an $\mathcal{R}$ -module by setting $M_j = M$ for $j \leq i$ and $M_j = 0$ for $j > i$. For $j \leq i$, $r \in R_j$ and $m \in M$ we then define $rm = d_{i-1} \dots d_j(r)m$. In particular, this makes the rings R_i themselves into $\mathcal{R}$ -modules.

Example 2.4. Let R_∞ be the direct limit of the sequence $\mathcal{R}$. Then similar to the previous example the ring homomorphisms $R_i \rightarrow R_\infty$ give R_∞ a natural $\mathcal{R}$ -module structure where R_∞ where the filtration is constant.

Lemma 2.5. Let $(M_j)_{j \in J}$ be a family of $\mathcal{R}$ -modules. Then the product of $(M_j)_{j \in J}$ in the category ${}_{\mathcal{R}}\mathbf{Mod}$ exists and is given by $\prod_{j \in J} M_j$ where $(\prod_{j \in J} M_j)_i = \prod_{j \in J} (M_j)_i$ and the $\mathcal{R}$ -structure is defined componentwise. Also the coproduct of $(M_j)_{j \in J}$ in the category ${}_{\mathcal{R}}\mathbf{Mod}$ exists and is the submodule of $\prod_{j \in J} M_j$ of elements with finite support, which we denote $\bigoplus_{j \in J} M_j$.

Definition 2.6. A *filtered set* is a set X together with a filtration $X_0 \supseteq X_1 \supseteq X_2 \supseteq \dots$. A map $f : X \rightarrow Y$ between filtered sets is a homomorphism if $f(X_i) \subseteq Y_i$ for all $i \geq 0$. The category of filtered sets is denoted $\mathbf{Set}_f$.

For X a filtered set, we define the degree function $\deg : X \rightarrow \mathbb{N} \cup \{\infty\}$ as $\deg(x) = \sup\{i \geq 0 \mid x \in X_i\}$.

Lemma 2.7. The forgetful functor ${}_{\mathcal{R}}\mathbf{Mod} \rightarrow \mathbf{Set}_f : M \mapsto M$ admits a left adjoint.

Proof. For a filtered set X consider the $\mathcal{R}$ -module

$$\mathcal{R}X = \bigoplus_{x \in X} R_{\deg(x)}x,$$

where R_i and R_∞ are given the $\mathcal{R}$ -module structure as in Example 2.3 and Example 2.4. One easily checks that this defines a functor that is left adjoint to the forgetful functor ${}_{\mathcal{R}}\mathbf{Mod} \rightarrow \mathbf{Set}_f$. $\square$

Definition 2.8. A homomorphism of filtered sets $f : X \rightarrow Y$ is

- (1) *degree preserving* if $f^{-1}(Y_i) = X_i$ for all $i \geq 0$. Note that this is equivalent to $\deg(m) = \deg(f(m))$ for all $x \in X$.
- (2) *a regular surjection* if the restriction $f_i : X_i \rightarrow Y_i$ is surjective for all $i \geq 0$.

Regular surjections are important since in the category $\mathcal{R}\mathbf{Mod}$ it is not true that every bijective homomorphism $f : M \rightarrow N$ is an isomorphism. Indeed, the inverse map f^{-1} is only a homomorphism of $\mathcal{R}$ -modules if $f^{-1}(N_i) \subseteq M_i$ for all $i \geq 0$, so if f is regular.

Lemma 2.9. *Let $f : M \rightarrow N$ be a homomorphism of $\mathcal{R}$ -modules. Then the following are equivalent:*

- (1) f is an isomorphism,
- (2) f is an injective regular surjection.

Definition 2.10. Let M, N, P be $\mathcal{R}$ -modules. A map $b : M \times N \rightarrow P$ is *bilinear* if

- (1) $b(M_i, N_j) \subseteq P_{i+j}$ for all $i, j \geq 0$,
- (2) The maps

$$b(-, n) : M \rightarrow P : m \mapsto b(m, n),$$

and

$$b(m, -) : N \rightarrow P : n \mapsto b(m, n),$$

are $\mathcal{R}$ -module homomorphisms for every $m \in M, n \in N$.

Proposition 2.11. *Let M, N be $\mathcal{R}$ -modules, there exists an $\mathcal{R}$ -module $M \otimes N$ and bilinear map $\otimes : M \times N \rightarrow M \otimes N$ such that for every $\mathcal{R}$ -module P and bilinear map $b : M \times N \rightarrow P$, there exists a unique $\mathcal{R}$ -module homomorphism $\bar{b} : M \otimes N \rightarrow P$ such that $b = \bar{b} \otimes$.*

Proof. Consider the set $X = M \times N$ with filtration $X_k = \bigcup_{i+j=k} M_i \times N_j$ and let $F = \mathcal{R}X$. We define $M \otimes N$ as F/I where I is the $\mathcal{R}$ -submodule of F generated by

$$\begin{aligned} & \{(m_1 + m_2, n) - (m_1, n) - (m_2, n) \mid m_1, m_2 \in M, n \in N\} \\ & \cup \{(m, n_1 + n_2) - (m, n_1) - (m, n_2) \mid m \in M, n_1, n_2 \in N\} \\ & \cup \{r(m, n) - (rm, n) \mid r \in R_i, m \in M_i, n \in N\} \\ & \cup \{r(m, n) - (m, rn) \mid r \in R_j, m \in M, n \in N_j\}. \end{aligned}$$

It is now easily seen that the universal property is satisfied. $\square$

As usual, for $m \in M, n \in N$ we will denote the image $\otimes(m, n)$ in $M \otimes N$ as $m \otimes n$. We can define $s_{M,N} : M \otimes N \rightarrow N \otimes M$ as the $\mathcal{R}$ -linear map such that $s_{M,N}(m \otimes n) = n \otimes m$. It is straightforward to verify the following proposition.

Proposition 2.12. *$(\mathcal{R}\mathbf{Mod}, \otimes, R_0, s)$ is symmetric monoidal additive category.*

Definition 2.13. Consider the symmetric monoidal additive category $(\mathcal{R}\mathbf{Mod}, \otimes, R_0, s)$,

- (1) an monoid in this category is called an $\mathcal{R}$ -algebra. We denote the category of $\mathcal{R}$ -algebras by $\mathcal{R}\mathbf{Alg}$.
- (2) a comonoid in this category is called a *co $\mathcal{R}$ -algebra*. The comultiplication and the counit of a co $\mathcal{R}$ -algebra will be denoted by Δ and ε . When evaluating Δ in specific elements, we will use Sweedler notation.
- (3) a Lie algebra object in this category is called a *Lie $\mathcal{R}$ -algebra*. We denote the category of Lie $\mathcal{R}$ -algebras by $\mathcal{R}\mathbf{Lie}$.
- (4) a Hopf monoid in this category is called a *Hopf $\mathcal{R}$ -algebra*. We denote the category of Hopf $\mathcal{R}$ -algebras by $\mathcal{R}\mathbf{Hopf}$. The antipode of a Hopf $\mathcal{R}$ -algebra we be denoted by S .

Example 2.14. Considering the original ring structure on the $\mathcal{R}$ -module R_0 makes it into an $\mathcal{R}$ -algebra.

Example 2.15. Let H be a filtered algebra (in the classical sense) over some field $\mathbb{F}$, then H is an $\overline{\mathbb{F}}$ -algebra. Note that every algebra can be made into a filtered algebra by choosing the constant filtration. The same construction applies to (filtered) Lie and Hopf-algebras over fields.

Example 2.16. Let M be an $\mathcal{R}$ -module. The ring $\text{End}_{\mathcal{R}}(M)$ of $\mathcal{R}$ -linear endomorphisms of M has a natural filtration given by

$$\text{End}_{\mathcal{R}}(M)_i = \{f \mid f(M_j) \subseteq M_{j+i} \text{ for all } j \geq 0\},$$

and a natural $\mathcal{R}$ -module structure given by $(rf)(m) = rf(m)$ for all $r \in R_i$, $f \in \text{End}_{\mathcal{R}}(M)_i$ and $m \in M$. This makes $\text{End}_{\mathcal{R}}(M)$ into an $\mathcal{R}$ -algebra.

Let H be a Hopf $\mathcal{R}$ -algebra H . An element $x \in H$ is *primitive* if $\Delta(x) = 1 \otimes x + x \otimes 1$. The set of primitive elements of H is denoted $\mathbb{P}(H)$. One easily verifies that $\mathbb{P}(H)$ is a sub $\mathcal{R}$ -module of H and that the operation $[x, y] = xy - yx$ makes $\mathbb{P}H$ into a Lie $\mathcal{R}$ -algebra. In this way we obtain a functor $\mathbb{P} : \mathcal{R}\mathbf{Hopf} \rightarrow \mathcal{R}\mathbf{Lie}$.

Given $M \in \mathcal{R}\mathbf{Mod}$, define $T^0(M) = R_0$ and $T^{i+1}(M) = M \otimes T^i(M)$. We consider $T(M) = \bigoplus_{i=0}^{\infty} T^i(M)$ as an $\mathcal{R}$ -algebra with the canonical multiplication and unit. One easily verifies the following property.

Proposition 2.17. *The functor $\mathcal{R}\mathbf{Mod} \rightarrow \mathcal{R}\mathbf{Alg} : M \mapsto T(M)$ is left adjoint to the functor $\mathbb{P} : \mathcal{R}\mathbf{Alg} \rightarrow \mathcal{R}\mathbf{Mod} : H \mapsto H$.*

It follows directly from Proposition 2.17 that there exists a unique $\mathcal{R}$ -algebra homomorphism $\Delta : T(M) \rightarrow T(M) \otimes T(M)$ satisfying $\Delta(m) = m \otimes 1 + 1 \otimes m$. Let $\varepsilon : T(M) \rightarrow R_0$ be the projection onto $T^0(M) = R_0$, and let $S : T(M) \rightarrow T(M)$ be the unique anti-automorphism such that $S(m) = -m$ for $m \in M$. Considering this extra structure on $T(M)$ is easily seen to make it into a (co-commutative) Hopf $\mathcal{R}$ -algebra which satisfies the following property.

Proposition 2.18. *The functor $\mathcal{R}\mathbf{Mod} \rightarrow \mathcal{R}\mathbf{Hopf} : M \mapsto T(M)$ is left adjoint to the functor $\mathbb{P} : \mathcal{R}\mathbf{Hopf} \rightarrow \mathcal{R}\mathbf{Mod} : H \mapsto \mathbb{P}(H)$.*

Proposition 2.19. *The functor $\mathbb{P} : \mathcal{R}\mathbf{Hopf} \rightarrow \mathcal{R}\mathbf{Lie}$ admits a left adjoint $\mathcal{U} : \mathcal{R}\mathbf{Lie} \rightarrow \mathcal{R}\mathbf{Hopf}$.*

Proof. Let $\mathfrak{g}$ be a Lie $\mathcal{R}$ -algebra and let I be the ideal of $T(\mathfrak{g})$ generated by

$$\{x \otimes y - y \otimes x - [x, y] \mid x, y \in L\}.$$

Then I is a Hopf ideal as $\Delta(x \otimes y - y \otimes x - [x, y]) \in I \otimes T(L) + T(L) \otimes I$ and $S(x \otimes y - y \otimes x - [x, y]) = y \otimes x - x \otimes y - [y, x] \in I$. We define $\mathcal{U}(\mathfrak{g}) = T(\mathfrak{g})/I$. This construction is easily seen to be functorial and from the universal property of $T(\mathfrak{g})$ it follows that also $\mathcal{U}(\mathfrak{g})$ satisfies the required universal property. $\square$

Example 2.20. Let M be an $\mathcal{R}$ -module and consider it as an abelian Lie $\mathcal{R}$ -algebra. We define $S(M) = \mathcal{U}(M)$, hence $S(M) = T(M)/I$ with I the ideal generated by $\{mn - nm \mid m, n \in M\}$. Equivalently we may see $S(M)$ as $\bigoplus_{i=0}^{\infty} S^i(M)$ where $S^0(M) = R_0$ and $S^i(M) = T^i(M)/M_i$ with M_i the $\mathcal{R}$ -submodule generated by

$$\{x_1 \otimes \cdots \otimes x_k - x_{\sigma(1)} \otimes \cdots \otimes x_{\sigma(k)} \mid x_1, \dots, x_k \in M, \sigma \in \Sigma_k\}.$$

3. DIVISIBLE SEQUENCES OF RINGS

Definition 3.1. A sequence of commutative rings $\mathcal{R}$ is said to be *divisible* if k is invertible in R_k for every $k > 0$.

Example 3.2. Let $\mathbb{Q}_0 = \mathbb{Z}$ and for $i \geq 1$ let $\mathbb{Q}_k$ denote the subring of $\mathbb{Q}$ consisting of fraction $\frac{n}{m}$ where all prime divisors of m are at most k . So

$$\mathbb{Q}_1 = \mathbb{Z}, \mathbb{Q}_2 = \{\frac{n}{2^k} \mid n \in \mathbb{Z}, k \in \mathbb{N}\}, \mathbb{Q}_3 = \{\frac{n}{2^k 3^l} \mid n \in \mathbb{Z}, k, l \in \mathbb{N}\}, \dots$$

Then $\mathcal{Q} : \mathbb{Q}_0 \rightarrow \mathbb{Q}_1 \rightarrow \mathbb{Q}_2 \rightarrow \dots$ is a divisible sequence of rings, where all homomorphisms are simply the inclusion.

It is easily seen that for $\mathcal{R}$ a divisible sequence, there exist unique homomorphisms $\iota_i : \mathbb{Q}_i \rightarrow R_i$ compatible with the maps of $\mathcal{Q}$ and $\mathcal{R}$. Hence, every $\mathcal{R}$ -module is a $\mathcal{Q}$ -module.

Example 3.3. Let R be a ring of non-zero characteristic and let p denote the smallest prime number dividing $\text{char } R$. Note that $1, \dots, p-1$ are invertible in R but p is not. We define $\tilde{R}$ as the sequence $R_0 = \dots = R_{p-1} = R$ and $0 = R_p = \dots$ with $d_i = \text{id}_R$ for $0 \leq i < p-1$. Then $\tilde{R}$ is a divisible sequence and $\tilde{R}$ -modules are precisely the R -modules M such that $M_p = 0$.

Definition 3.4. A filtered set X is *connected* if $X_0 = X_1$. An $\mathcal{R}$ -module or Lie $\mathcal{R}$ -algebra is *connected* if its underlying filtered set is connected. A Hopf $\mathcal{R}$ -algebra H is *connected* if $H_1 = \ker \varepsilon$.

We are now able to prove the Poincaré-Birkhoff-Witt theorem in our setting.

Theorem 3.5. Let $\mathcal{R}$ be divisible and $\mathfrak{g}$ a connected Lie $\mathcal{R}$ -algebra. Then there exists an isomorphism of co $\mathcal{R}$ -algebras $\phi : S(\mathfrak{g}) \rightarrow \mathcal{U}(\mathfrak{g})$ and ϕ restricted to $\mathfrak{g} = S^1(\mathfrak{g})$ is the canonical map.

Proof. We denote the symmetric group on $\{1, \dots, k\}$ as Σ_k . For readability, we also write $S = S(\mathfrak{g})$, $T^i = T^i(\mathfrak{g})$ and $T = T(\mathfrak{g})$. Consider the $\mathcal{R}$ -linear homomorphism

$$\psi : S \rightarrow T : x_1 \cdots x_k \mapsto \frac{1}{k!} \sum_{\sigma \in \Sigma_k} x_{\sigma(1)} \cdots x_{\sigma(k)},$$

which is well-defined as any product of k elements in L is contained in T_k and k is invertible in $\mathcal{R}_k$. We also use the notation $\sigma(x_1 \cdots x_k) = x_{\sigma(1)} \cdots x_{\sigma(k)}$, and we extend this linearly to obtain an action of Σ_k on T^k .

Let I denote the ideal of $T(\mathfrak{g})$ generated by $\{xy - yx - [x, y] \mid x, y \in L\}$. We claim that $T = \psi(S) \oplus I$, which indeed implies that ψ yields an isomorphism $S \rightarrow T/I \cong \mathcal{U}(\mathfrak{g})$. We inductively define maps $\rho_n : T^n \rightarrow T$, as

$$\rho_0(r) = r, \quad \rho_n(w) = \frac{1}{n!} \sum_{\sigma \in \Sigma_n} \sigma(w) + f_\sigma(w)$$

for $r \in R_0$ and $w \in T^n(\mathfrak{g})$. For $\sigma \in \Sigma_k$, we define the $\mathcal{R}$ -linear homomorphism $f_\sigma : T^n \rightarrow T^{n-1}$ in the following way:

- (1) $f_{\text{id}}(w) = 0$,
- (2) for an adjacent transposition $\sigma = (j \ j+1) \in \Sigma_k$,

$$f_\sigma(x_1 \dots x_n) = \rho_{n-1}(x_1 \dots x_{j-1} [x_j, x_{j+1}] x_{j+2} \dots x_n).$$

- (3) For each $\sigma \in \Sigma_n$ we fix a decomposition $\sigma = \sigma_{l(\sigma)} \cdots \sigma_1$ where σ_i are adjacent transpositions. Note that one can for example always take the unique decomposition of minimal length which is lexicographically minimal, so we do not require the axiom of choice or any variation thereof. We define $p_0^\sigma = \text{id}$, $p_{i+1}^\sigma = \sigma_{i+1} p_i^\sigma$, and $f(\sigma, w) = \sum_{i=1}^{l(\sigma)} f_{\sigma_i}(p_{i-1}^\sigma(w))$.

Combining these maps ρ_n we obtain an $\mathcal{R}$ -module endomorphism $\rho : T \rightarrow T$.

We claim that $\rho(T) \subseteq \psi(S)$, $(\text{id} - \rho)(T) \subseteq I$ and $\rho^2 = \rho$, which is sufficient to conclude that $T = \psi(S) \oplus I$. The first inclusion follows inductively from the definition. For the second inclusion, clearly $(\text{id} - \rho)(T^n) = 0$ for $n \in \{0, 1\}$. For

$n > 1$, $\sigma \in \Sigma_n$ with decomposition $\sigma_{l(\sigma)} \cdots \sigma_1$ and $w \in T^n$ we find

$$\begin{aligned} w - \sigma(w) - f(\sigma, w) &= p_0^\sigma(w) - p_{l(\sigma)}^\sigma(w) - \sum_{i=1}^{l(\sigma)} f_{\sigma_i}(p_{i-1}(w)) \\ &= \sum_{i=1}^{l(\sigma)} p_{i-1}^\sigma(w) - p_i^\sigma(w) - f_{\sigma_i}(p_{i-1}(w)) \in I, \end{aligned}$$

as $w - \tau(w) - f_\tau(w) \in I$ for any adjacent transposition τ . Hence, $(\text{id} - \rho)(w) \in I$.

As ρ behaves as the identity on T^0 and T^1 , clearly $\rho^2 = \rho$ on $T^0 \oplus T^1$. Let $n > 1$, for any adjacent transposition $\sigma \in \Sigma_n$ and $w \in T^n$, we find that $f_\sigma(w) = -f_\sigma(\sigma(w))$ hence

$$\sum_{\tau \in \Sigma_n} f_\sigma(\tau(w)) = \frac{1}{2} \sum_{\tau \in \Sigma_n} f_\sigma(\tau(w)) + f_\sigma(\tau(w)) = \frac{1}{2} \sum_{\tau \in \Sigma_n} f_\sigma(\tau(w)) + f_\sigma(\sigma\tau(w)) = 0,$$

which consequently holds for any $\sigma \in \Sigma_n$. We conclude that

$$\rho_n(\psi(w)) = \frac{1}{n!} \sum_{\sigma \in \Sigma_n} \sigma(\psi(w)) + f_\sigma(\psi(w)) = \frac{1}{(n!)^2} \sum_{\sigma, \tau \in \Sigma_n} \sigma\tau(w) + f_\sigma(\tau(w)) = \psi(w).$$

Which proves that $\rho^2 = \rho$. From the above we conclude that $T = \psi(S) \oplus I$, thus ψ yields an isomorphism of $\mathcal{R}$ -modules $\phi : S \rightarrow \mathcal{U}(\mathfrak{g})$.

We conclude by proving that ϕ is a co $\mathcal{R}$ -algebra homomorphism. Clearly ϕ is compatible with the co-unit maps. Also, $(\phi \otimes \phi)\Delta_S(r) = \Delta_{\mathcal{U}(\mathfrak{g})}\phi(r)$ for $r \in T^0$. We will use the following convention when working in some algebra A : for $I \subseteq \{1, \dots, n\}$ with $|I| = k$ and elements $a_i \in A$, $i \in I$, then by $\prod_{i \in I} a_i$ we mean the product $a_{i_1} \cdots a_{i_k}$ where $i_1 < \dots < i_k$. By I^c we mean $\{1, \dots, n\} \setminus I$. For $x_1, \dots, x_n \in \mathfrak{g}$,

$$\begin{aligned} (\phi \otimes \phi)(\Delta_S(x_1 \cdots x_n)) &= \sum_{I \subseteq \{1, \dots, n\}} \phi\left(\prod_{i \in I} x_i\right) \otimes \phi\left(\prod_{i \notin I} x_i\right) \\ &= \sum_{I \subseteq \{1, \dots, n\}} \frac{1}{|I|!} \frac{1}{(n - |I|)!} \left(\sum_{\sigma \in \Sigma_I} \prod_{i \in I} x_{\sigma(i)} \right) \otimes \left(\sum_{\tau \in \Sigma_{I^c}} \prod_{j \in I^c} x_{\tau(j)} \right) \\ &= \frac{1}{n!} \sum_{k=1}^n \sum_{\substack{I \subseteq \{1, \dots, n\}, \\ |I|=k}} \binom{n}{k} \left(\sum_{\sigma \in \Sigma_I} \prod_{i \in I} x_{\sigma(i)} \right) \otimes \left(\sum_{\tau \in \Sigma_{I^c}} \prod_{j \in I^c} x_{\tau(j)} \right), \end{aligned}$$

which equals $\frac{1}{n!}$ times the sum of $\binom{n}{k}(x_{i_1} \cdots x_{i_k}) \otimes (x_{j_1} \cdots x_{j_{n-k}})$ with $1 \leq k \leq n$ and $(i_1, \dots, i_k, j_1, \dots, j_{n-k})$ all possible permutations of $(1, \dots, n)$. On the other hand,

$$\begin{aligned} \Delta_{\mathcal{U}(\mathfrak{g})}(\phi(x_1 \cdots x_n)) &= \frac{1}{n!} \sum_{\sigma \in \Sigma_n} \prod_{i=1}^n \Delta(x_{\sigma(i)}) \\ &= \frac{1}{n!} \sum_{\sigma \in \Sigma_n} \prod_{i=1}^n (x_{\sigma(i)} \otimes 1 + 1 \otimes x_{\sigma(i)}) \\ &= \frac{1}{n!} \sum_{\sigma \in \Sigma_n} \sum_{I \subseteq \{1, \dots, n\}} \left(\prod_{i \in I} x_{\sigma(i)} \right) \otimes \left(\prod_{i \in I^c} x_{\sigma(i)} \right). \end{aligned}$$

For $(i_1, \dots, i_k, j_1, \dots, j_{n-k})$ a permutation of $(1, \dots, n)$ and any choice of I with $|I| = k$, there is precisely one permutation $\sigma \in \Sigma_n$ such that $(\prod_{i' \in I} x_{\sigma(i')}) \otimes (\prod_{j' \in I^c} x_{\sigma(j')}) = (x_{i_1} \cdots x_{i_k}) \otimes (x_{j_1} \cdots x_{j_{n-k}})$. So the summand $(x_{i_1} \cdots x_{i_k}) \otimes (x_{j_1} \cdots x_{j_{n-k}})$ appears precisely $\binom{n}{k}$ times in the latter sum, which proves that $(\phi \otimes \phi)\Delta_S = \Delta_{\mathcal{U}(\mathfrak{g})}\phi$. $\square$

4. COMPLETE HOPF $\mathcal{R}$ -ALGEBRAS

We can naturally consider an $\mathcal{R}$ -module M as a topological space where the base of the topology is given by $\{m + M_i \mid m \in M, i \geq 0\}$. With this topology, homomorphism of $\mathcal{R}$ -modules are automatically continuous. Note that if $M_i = 0$ for some i , then we obtain the discrete topology on M .

Definition 4.1. An $\mathcal{R}$ -module M is *complete* if $\bigcap_{i=0}^{\infty} M_i = 0$ and it is complete with respect to the above topology. The full subcategory of $\mathcal{R}\mathbf{Mod}$ consisting of complete $\mathcal{R}$ -modules is denoted $\mathcal{R}\mathbf{Mod}^c$.

One easily verifies that

$$\hat{M} = \left\{ (m_1 + M_1, m_2 + M_2, \dots) \in \prod_{i=1}^{\infty} M/M_i \mid m_{i+1} + M_i = m_i + M_i \text{ for } i \geq 0 \right\},$$

is a sub $\mathcal{R}$ -module of $\prod_{i=1}^{\infty} M/M_i$ which is complete and the image of the $\mathcal{R}$ -module homomorphism $M \rightarrow \hat{M} : m \mapsto (m + M_1, m + M_2, \dots)$ is dense in $\hat{M}$. Moreover, if M is complete then $M \cong \hat{M}$. This construction can easily be extended to a functor $\hat{\cdot} : \mathcal{R}\mathbf{Mod} \rightarrow \mathcal{R}\mathbf{Mod}^c$ in the natural way. For two complete $\mathcal{R}$ -modules M and N , we define $M \hat{\otimes} N$ as $\widehat{M \otimes N}$.

Proposition 4.2. *The category $(\mathcal{R}\mathbf{Mod}^c, R_0, \hat{\otimes}, \hat{s})$ is a symmetric monoidal additive category and the functor $\hat{\cdot} : \mathcal{R}\mathbf{Mod} \rightarrow \mathcal{R}\mathbf{Mod}^c$ is symmetric monoidal additive.*

Proof. We will prove that for any $M, N \in \mathcal{R}\mathbf{Mod}$, $\widehat{M \otimes N}$ is canonically isomorphic to $\hat{M} \hat{\otimes} \hat{N}$, as this is easily seen to imply the associativity of $\hat{\otimes}$. Note that we have a natural homomorphism $f_1 : M \otimes N \rightarrow \hat{M} \otimes \hat{N}$, hence also a natural homomorphism $f_2 : \widehat{M \otimes N} \rightarrow \hat{M} \hat{\otimes} \hat{N}$.

Define $\hat{M} \times \hat{N} \rightarrow \widehat{M \otimes N}$ by mapping $((m_i)_i, (n_i)_i)$ to $((m_i \otimes n_i))_i$ for $(m_i)_i \in \hat{M}$ and $(n_i)_i \in \hat{N}$ which is easily seen to be a well-defined bilinear map of $\mathcal{R}$ -modules. From the universal property we get a homomorphism $g_1 : \hat{M} \otimes \hat{N} \rightarrow \widehat{M \otimes N}$ and thus also a homomorphism $g_2 : \hat{M} \hat{\otimes} \hat{N} \rightarrow \widehat{M \otimes N}$.

Note that the canonical image of $M \otimes N$ into $\widehat{M \otimes N}$ is dense. Also its canonical image into $\hat{M} \otimes \hat{N}$ is dense, hence the same is true for its canonical image in $\hat{M} \hat{\otimes} \hat{N}$. Moreover, on the canonical images of $M \otimes N$ in $\hat{M} \otimes \hat{N}$ and $\widehat{M \otimes N}$, g_2 and f_2 are mutually inverse. Because these canonical images are dense, it follows that f_2 and g_2 are mutually inverse and the desired canonical isomorphism follows.

It is clear that the $\mathcal{R}$ -module R_0 is complete and from the definition of $\hat{\otimes}$ it is clear that $\hat{\cdot}$ yields a monoidal functor. Also the additivity is clear of both the category and the functor follow easily. $\square$

Definition 4.3. Consider the symmetric monoidal additive category $(\mathcal{R}\mathbf{Mod}^c, \hat{\otimes}, R_0, \hat{s})$,

- (1) a monoid in this category is called a *complete $\mathcal{R}$ -algebra*. We denote the category of complete $\mathcal{R}$ -algebras by $\mathcal{R}\mathbf{Alg}^c$,
- (2) a comonoid in this category is called a *co $\mathcal{R}$ -algebra*,
- (3) a Lie algebra object in this category is called a *complete Lie $\mathcal{R}$ -algebra*. We denote the category of complete Lie $\mathcal{R}$ -algebras by $\mathcal{R}\mathbf{Lie}^c$,
- (4) a Hopf monoid in this category is called a *complete Hopf $\mathcal{R}$ -algebra*. We denote the category of complete Hopf $\mathcal{R}$ -algebras by $\mathcal{R}\mathbf{Hopf}^c$.

In particular, $\hat{\cdot}$ yields a functor $\mathcal{R}\mathbf{Hopf} \rightarrow \mathcal{R}\mathbf{Hopf}^c$ and we define $\hat{\mathcal{U}} : \mathcal{R}\mathbf{Lie}^c \rightarrow \mathcal{R}\mathbf{Hopf}^c$ as the composition of $\hat{\cdot}$ after $\mathcal{U}$.

We also define $\hat{S}(M)$ as $\widehat{S(M)}$. If M is connected one easily verifies that $\hat{S}(M) \cong \prod_{k=0}^{\infty} \hat{S}^k(M)$ where $\hat{S}^k(M) = \widehat{S^k(M)}$. The following version of the Poincaré-Birkhoff-Witt theorem is now a straightforward corollary of Theorem 3.5.

Theorem 4.4. *Let $\mathcal{R}$ be divisible and $\mathfrak{g}$ a complete connected Lie $\mathcal{R}$ -algebra. Then there exists an isomorphism of complete co $\mathcal{R}$ -algebras $\phi : \hat{S}(\mathfrak{g}) \rightarrow \hat{\mathcal{U}}(\mathfrak{g})$ and ϕ restricted to $L = \hat{S}^1(\mathfrak{g})$ is the canonical map.*

For a complete Hopf $\mathcal{R}$ -algebra H , an element $x \in H$ is *primitive* if $\Delta(x) = 1 \hat{\otimes} x + x \hat{\otimes} 1$ and we denote the set of primitive elements as $\mathbb{P}(H)$. As the primitive elements are precisely the kernel of the $\mathcal{R}$ -linear map $H \rightarrow H \hat{\otimes} H$ mapping $h \in H$ to $\Delta(h) - 1 \hat{\otimes} h - h \hat{\otimes} 1$, we find that $\mathbb{P}(H)$ is closed hence complete. We therefore obtain a functor $\mathbb{P} : \mathcal{R}\mathbf{Hopf}^c \rightarrow \mathcal{R}\mathbf{Lie}^c$.

Lemma 4.5. *Let $\mathcal{R}$ be divisible and let M be a complete connected $\mathcal{R}$ -module, then $\mathbb{P} \hat{S}(M) = \hat{S}^1(M)$.*

Proof. The elements of $\hat{S}^1(M)$ are clearly primitive. Let $\pi : \hat{S}(M) \rightarrow \hat{S}(M)$ be the projection onto $\hat{S}^1(M)$ and $\theta : \prod_{i=1}^{\infty} \hat{S}^i(M) \rightarrow \hat{S}(M)$ the $\mathcal{R}$ -module homomorphism which maps $w \in \hat{S}^i(M)$ to $\frac{1}{i}w$, where the connectedness of M is crucial to assure that this is well-defined. At last, we define

$$\psi : \prod_{i=1}^{\infty} \hat{S}^i(M) \rightarrow \prod_{i=1}^{\infty} \hat{S}^i(M) : h \mapsto \theta(h_{(1)}\pi(h_{(2)})).$$

Then $\psi(a) = a$ for any $a \in \hat{S}^i(M)$ with $i > 0$, hence $\psi = \text{id}$. Also $\psi(x) = \pi(x)$ for a primitive element $x \in H$. In particular, for any primitive element $x = \sum_{k=1}^{\infty} x_k$ with $x_k \in \hat{S}^k(M)$, we find that $x_1 = \sum_{k=1}^{\infty} x_k$, hence we find that $x \in \hat{S}^1(M)$. $\square$

Lemma 4.6. *Let $\mathcal{R}$ be divisible, $\mathfrak{g}$ be a complete connected Lie $\mathcal{R}$ -algebra, C a complete co $\mathcal{R}$ -algebra and $f : \hat{\mathcal{U}}(\mathfrak{g}) \rightarrow C$ a homomorphism of co $\mathcal{R}$ -algebras. If the restriction $f : \mathbb{P}\hat{\mathcal{U}}(\mathfrak{g}) \rightarrow \mathbb{P}C$ is degree preserving, then f is injective.*

Proof. By Theorem 4.4 we may replace $\hat{\mathcal{U}}(\mathfrak{g})$ by $\hat{S}(\mathfrak{g})$, recall that $\mathbb{P}\hat{S}(\mathfrak{g}) = \hat{S}^1(\mathfrak{g})$ by Lemma 4.5. We write $\hat{S} = \hat{S}(\mathfrak{g})$ and $\hat{S}^i = \hat{S}^i(\mathfrak{g})$. Assume that $f : \mathbb{P}\hat{S} \rightarrow C$ is degree preserving but that f is non-injective. Let $I = \ker f$, hence it is a non-trivial coideal of $\hat{S}$.

Let π, θ, ψ be as defined in the proof of Lemma 4.5 and let $\mu : \hat{S} \otimes \hat{S} \rightarrow \hat{S}(\mathfrak{g})$ denote the multiplication. Let d be maximal such that $I \subseteq \hat{S}_d$, so in particular $d \geq 1$. Then

$$I = \psi(I) \subseteq \mu(\pi \hat{\otimes} \text{id})(I \hat{\otimes} \hat{S} + \hat{S} \hat{\otimes} I) \subseteq \mu(\hat{S}_d^1 \hat{\otimes} \hat{S} + \hat{S}^1 \hat{\otimes} \hat{S}_d) \subseteq \hat{S}_d^1 + S_{d+1}.$$

Take $a \in I$ any element of degree d , we can write $a = a_1 + a_2$ with $a_1 \in \hat{S}_d^1$ and $a_2 \in S_{d+1}$. As $\deg(a) \geq \min\{\deg(a_1), \deg(a_2)\}$ and $\deg(a_2) > d$, we find that $\deg(a_1) = d$. However, we now find

$$\deg(a_2) > \deg(a_1) = \deg(f(a_1)) = \deg(-f(a_2)) = \deg(f(a_2)),$$

which is impossible. $\square$

A complete $\mathcal{R}$ -algebra A is *generated* by a subset $X \subseteq A$ (which we consider as a filtered set by setting $X_i = X \cap A_i$) if for each $i \geq 0$, A_i is the closed sub $\mathcal{R}$ -module generated by

$$\{x_1 \cdots x_r \mid k_1 + \dots + k_r = i, x_i \in X_{k_i}\},$$

where for $i = 0$, we also include the empty product 1 in this set. A complete Hopf $\mathcal{R}$ -algebra H is *primitively generated* if $\mathbb{P}(H)$ generates the complete $\mathcal{R}$ -algebra H .

The full subcategory of $\mathcal{R}\mathbf{Hopf}^c$ consisting of primitively generated complete connected Hopf $\mathcal{R}$ -algebras is denoted $\mathcal{R}\mathbf{Hopf}_{pg}^{c,c}$. Clearly $\hat{T}(M)$ is primitively generated and connected, and if $f : \hat{T}(M) \rightarrow H$ is a regular surjective homomorphism of complete Hopf $\mathcal{R}$ -algebras, then also H is primitively generated and connected.

The full subcategory of $\mathcal{R}\mathbf{Lie}^c$ consisting of complete connected Lie $\mathcal{R}$ -algebras is denoted $\mathcal{R}\mathbf{Lie}^{c,c}$. We are now ready to prove the Minor-Moore theorem in our setting.

Theorem 4.7. *Let $\mathcal{R}$ be divisible, then the functors $\hat{\mathcal{U}}$ and $\mathbb{P}$ define an equivalence of the categories $\mathcal{R}\mathbf{Lie}^{c,c}$ and $\mathcal{R}\mathbf{Hopf}_{pg}^{c,c}$.*

Proof. One easily verifies that we can indeed restrict the functors $\hat{\mathcal{U}}$ and $\mathbb{P}$ to $\mathcal{R}\mathbf{Lie}^{c,c}$ and $\mathcal{R}\mathbf{Hopf}_{pg}^{c,c}$.

Let $\mathfrak{g} \in \mathcal{R}\mathbf{Lie}^{c,c}$, then the natural homomorphism $f : L \rightarrow \mathbb{P}\hat{\mathcal{U}}L$ is an isomorphism by Theorem 4.4 and Lemma 4.5.

Let H be a primitively generated complete connected Hopf $\mathcal{R}$ -algebra, then the canonical homomorphism of Hopf $\mathcal{R}$ -algebras $f : \hat{\mathcal{U}}\mathbb{P}(H) \rightarrow H$ is a surjection which is regular as both Hopf $\mathcal{R}$ -algebras are primitively generated. The injectivity follows from Lemma 4.6 as f restricts to the isomorphism $\mathbb{P}(\mathfrak{g}) \rightarrow \mathbb{P}\hat{\mathcal{U}}(\mathfrak{g})$. We conclude that f is injective and thus it is an isomorphism by Lemma 2.9. $\square$

5. THE LAZARD CORRESPONDENCE FOR LIE $\mathcal{R}$ -ALGEBRAS AND LIE $\mathcal{R}$ -GROUPS

Definition 5.1. Let R be a commutative ring. An R -group is a group G with a map $R \times G \rightarrow G : (r, g) \mapsto g^r$ satisfying for all $r, s \in R, g, h \in G$

- (1) $g^1 = g$,
- (2) $g^{r+s} = g^r g^s$,
- (3) $g^{rs} = (g^r)^s$,
- (4) $hg^r h^{-1} = (hgh^{-1})^r$ (where h^{-1} denotes the inverse of h).

A homomorphism of $f : G \rightarrow H$ of R -groups is a group homomorphism such that moreover $f(g^r) = f(g)^r$ for all $g \in G, r \in R$.

Note that R -groups were also defined in [41], but their definition has the extra restriction that the Hall-Petresco formula should hold. However, we will impose a similar condition for $\mathcal{R}$ -groups later on in the form of (G3). We can make the following easy observations in an R -group G , for $r \in R, g \in G$:

- (1) $g^0 = g^0 g^0$ hence $g^0 = 1$,
- (2) $1 = g^0 = g^r g^{-r}$ hence g^{-r} is the inverse of g^r ,
- (3) for every $n = 1 + \dots + 1 \in R$, the element g^n is necessarily the n -th power of g in G .

In particular, every group has a unique $\mathbb{Z}$ -group structure.

Proposition 5.2. *Let G be a group and let $\mathbb{Q}_k$ be as in Example 3.2 Then G is $\mathbb{Q}_k$ -group if and only if for every $g \in G$ and every prime $p \leq k$ there exists a unique $h \in G$ such that $h^p = g$. Moreover, any group has at most one possible $\mathbb{Q}$ -group structure.*

Proof. Assume that G is a $\mathbb{Q}_k$ -group and $p \leq k$ then clearly $(g^{\frac{1}{p}})^p = g$. If h is any element such that $h^p = g$, then $h = (h^p)^{\frac{1}{p}} = g^{\frac{1}{p}}$, which proves the uniqueness. Conversely, assume that for every g and $p \leq k$ there exists a unique h such that $h^p = g$. Then in particular for every n whose prime divisors are at most k , we can find a unique h such that $h^n = g$, we denote this element $\sqrt[n]{g}$. The unique $\mathbb{Q}_k$ -group structure should necessarily be given by $g^{\frac{a}{n}} := \sqrt[n]{g^a}$. One easily verifies that G indeed becomes a $\mathbb{Q}_k$ -group. $\square$

Definition 5.3. A *filtered group* is a group G together with a descending filtration of normal subgroups $G = G_0 \supseteq G_1 \supseteq \dots$ such that $[G_i, G_j] \subseteq G_{i+j}$ for all $i, j \geq 1$, where $[-, -]$ denotes the group theoretic commutator. A homomorphism of filtered groups $f : G \rightarrow H$ is a group homomorphism such that $f(G_i) \subseteq H_i$.

A filtered group G is *complete* if $\bigcap G_i = \{e\}$ and it is complete with respect to the topology generated by the base $\{gG_i \mid g \in G, i \geq 1\}$. A filtered group is *connected* if $G_0 = G_1$.

Definition 5.4. An $\mathcal{R}$ -group is a filtered group G such that

- (G1) G_i is an R_i -group for every $i \geq 1$,
- (G2) for any $i \geq 0$, $r \in R_i$ and $g \in G_i$, $g^r = g^{d_i(r)}$,
- (G3) $(gh)^r G_{i+1} = g^r h^r G_{i+1}$, for all $i \geq j \geq 0$, $g \in G_i$, $h \in G_j$, $r \in R_j$.

A homomorphism of $\mathcal{R}$ -groups $f : G \rightarrow H$ is a homomorphism of filtered groups such that $f : G_i \rightarrow H_i$ is a R_i -group homomorphism for all $i \geq 0$. The category of complete connected $\mathcal{R}$ -group is denoted $\mathcal{R}\mathbf{Gr}^{c,c}$.

Example 5.5. Let G be a group and define $G_0 = G_1 = G$ and $G_{i+1} = [G, G_i]$ for $i \geq 1$. Then G has a unique $\mathbb{Z}$ -group structure.

Proposition 5.6. Let G be a filtered group. Then G is $\mathcal{Q}$ -group if and only if for every $i \geq 1$, every $g \in G_i$ and every prime $p \leq i$, there exists a unique element $h \in G_i$ such that $h^p = g$.

If moreover G is torsion, then G is a $\mathcal{Q}$ -group if and only if $\gcd(|g|, i) = 1$ for all $i \geq 1$, $g \in G_i$, where $|g|$ denotes the order of g . Moreover, every subgroup of G is a sub $\mathcal{R}$ -group and the automorphisms of G as a $\mathcal{Q}$ -group coincide with the automorphisms of G as a filtered group.

Proof. The first part follows directly from Proposition 5.2 and an easy verification of (G3).

Let $g \in G_i$, if $|g|$ is coprime to p as then by Bezout's theorem there exists some r, s such that $1 = rp + s|g|$, hence $(g^r)^p = g$ and $g^r \in G_i$. If $h^p = g$ for some other $h \in G_i$, then since $|h|$ is coprime to p , we find $|g| = |h^p| = |h|$ thus $h = (h^p)^r = g^r$. Conversely, if for some $i \geq 1$, $p \leq i$ and $g \in G_i$, p divides $|g|$, then without loss of generality $|g| = p$. If $h \in G_i$ is such $h^p = g$, then also $(hg)^p$ hence the uniqueness is not satisfied. $\square$

Let A be a complete $\mathcal{R}$ -algebra with $\mathcal{R}$ divisible. Then $\exp(a) := \sum_{i=0}^{\infty} \frac{1}{i!} a^i$ is a well-defined element of $1 + A_1$ for any $a \in A_1$. Also, $\log(a) := \sum_{i=1}^{\infty} \frac{1}{i} (-1)^{i+1} (a-1)^i$ is a well-defined element of A_1 for any $a \in 1 + A_1$.

For a complete Hopf $\mathcal{R}$ -algebra H an element $g \in H$ is *grouplike* if $\Delta(g) = g \hat{\otimes} g$. The set of grouplike elements of H is denoted $\mathbb{G}(H)$. Note that this immediately implies that $gS(g) = S(g)g = 1$ hence $\mathbb{G}(H)$ is a group.

Theorem 5.7. Let $\mathcal{R}$ be divisible and H be a complete $\mathcal{R}$ -algebra, then $\exp$ and $\log$ yield bijections between the sets $1 + H_i$ and H_i for all $i \geq 1$. If moreover H is a complete connected Hopf $\mathcal{R}$ -algebra then this bijection restricts to a bijection between $\mathbb{G}(H)$ and $\mathbb{P}(H)$.

Proof. The first part of the statement is straightforward to verify. Now assume that H is a complete Hopf $\mathcal{R}$ -algebra and let $g \in \mathbb{G}(H)$. As Δ is an $\mathcal{R}$ -algebra morphism, we find for any grouplike element $g \in H$ that $\Delta(\log(g)) = \log(\Delta(g)) = \log(g \hat{\otimes} g)$, where on the right hand side we consider the logarithm of $\Delta(g)$ in $H \hat{\otimes} H$. The usual rule for a logarithm of a commuting product, which is readily verified to also hold in our algebraic setting, now yields

$$\Delta(\log(g)) = \log((g \hat{\otimes} 1)(1 \hat{\otimes} g)) = \log(g \hat{\otimes} 1) + \log(1 \hat{\otimes} g) = \log(g) \hat{\otimes} 1 + 1 \hat{\otimes} \log(g).$$

Hence, we find that $\log(g) \in \mathbb{P}(A)$.

If we start with a primitive element $x \in \mathbb{P}(A)$, we use an argument similar to the one above to get

$$\begin{aligned}\Delta(\exp(x)) &= \exp(x \hat{\otimes} 1 + 1 \hat{\otimes} x) = \exp(x \hat{\otimes} 1) \exp(1 \hat{\otimes} x) \\ &= (\exp(x) \hat{\otimes} 1)(1 \hat{\otimes} \exp(x)) = \exp(x) \hat{\otimes} \exp(x).\end{aligned}$$

We conclude that $\exp(x) \in \mathbb{G}(H)$. $\square$

Lemma 5.8. *Let $\mathcal{R}$ be divisible and let H be a complete connected Hopf $\mathcal{R}$ -algebra. Let $i, j \geq 1$, $x \in H_i$, $y \in H_j$, then*

$$\log(\exp(x) \exp(y) \exp(x)^{-1} \exp(y)^{-1}) + H_{i+j+1} = [x, y] + H_{i+j+1}.$$

Proof. Note that every product of elements of $\{x, y\}$ of length at least three containing at least one occurrence of x and y is contained in H_{i+j+1} .

In particular, $x \exp(y) + H_{i+j+1} = x + xy + H_{i+j+1}$ and $x^k \exp(y) + H_{i+j+1} = x^k + H_{i+j+1}$ for $k > 1$. So

$$\begin{aligned}\exp(x) \exp(y) + H_{i+j+1} &= (1 + x + \frac{1}{2}x^2 + \frac{1}{3!}x^3 + \dots) \exp(y) + H_{i+j+1} \\ &= \exp(y) + x + xy + \frac{1}{2}x^2 + \frac{1}{3!}x^3 + \dots + H_{i+j+1} \\ &= \exp(x) + \exp(y) + xy - 1 + H_{i+j+1}\end{aligned}$$

and in the above x can be exchanged for $-x$ and y for $-y$. This means that

$$\begin{aligned}\exp(x) \exp(y) \exp(-x) \exp(-y) + H_{i+j+1} &= (\exp(x) + \exp(y) + xy - 1)(\exp(-x) + \exp(-y) + xy - 1) + H_{i+j+1} \\ &= 1 + \exp(x) \exp(-y) + xy - \exp(x) + \exp(y) \exp(-x) + 1 + xy - \exp(y) \\ &\quad + xy + xy - xy - \exp(-x) - \exp(-y) - xy + 1 + H_{i+j+1} \\ &= 1 + xy - yx + H_{i+j+1} \\ &= \exp([x, y]) + H_{i+j+1}\end{aligned}$$

$\square$

Proposition 5.9. *Let $\mathcal{R}$ be divisible and let H be a complete connected Hopf $\mathcal{R}$ -algebra. Then $\mathbb{G}(H)$ is a complete $\mathcal{R}$ -group if we set $\mathbb{G}(H)_i = \mathbb{G}(H) \cap (1 + H_i)$ and $g^r = \exp(r \log(g))$ for $g \in \mathbb{G}(H)_i$ and $r \in R_i$. Moreover, $\exp(x + \mathbb{P}(H)_i) = \exp(x) \mathbb{G}(H)_i$, for all $i \geq 1$ hence $\exp$ and $\log$ are homeomorphisms between $\mathbb{P}(H)$ and $\mathbb{G}(H)$.*

Proof. For simplicity let us denote $G = \mathbb{G}(H)$. Clearly $1 + H_i$ is closed under multiplication, hence G_i is closed under multiplication. It is clear from the definition that if $g \in G_i$, then also $g^r \in G_i$ for $r \in R_i$, so in particular $g^{-1} \in G_i$. One easily sees that (G1) and (G2) hold.

As $\exp(x + H_i) \subseteq \exp(x) + H_i$ and $\log(\exp(x) + H_i) \subseteq x + H_i$ for any $x \in \mathbb{P}(H)$, we find indeed that $\exp(x + \mathbb{P}(H)_i) = \exp(x)G_i$. We conclude that $\exp$ and $\log$ are homeomorphisms between $\mathbb{P}(H)$ and G so in particular G is complete.

The fact that $[G_i, G_j] \subseteq G_{i+j}$ follows from Lemma 5.8.

We conclude by proving that (G3) holds. Let $i \geq j \geq 0$, $g \in G_i$, $h \in G_j$ and $r \in R_j$, because g and h commute in H/H_{i+1} we find

$$(gh)^r + H_{i+1} = \exp(r \log(g) + r \log(h)) + H_{i+1} = g^r h^r + H_{i+1}. \quad \square$$

Lemma 5.10. *Let G be an $\mathcal{R}$ -group generated by $X \subseteq G$, then $|G| \leq \max\{|X|, \alpha, \aleph_0\}$ where $\alpha = |\bigcup_{i=1}^{\infty} R_i|$.*

Proof. Let $A_0 = X$ and define inductively B_i as the group generated by A_i and $A_{i+1} = \{b^r \mid b \in B_i, r \in R_{\deg b}\}$. If we let H be the union of the chain $A_0 \subseteq B_0 \subseteq A_1 \subseteq B_1 \subseteq \dots$, then H is a sub $\mathcal{R}$ -group of G . Note that $|B_i| \leq \max\{|A_i|, \aleph_0\}$. Also,

$|A_{i+1}| \leq \alpha|B_i|$. In particular, $|A_i| \leq \max\{|X|, \alpha, \aleph_0\}$, hence $|H| \leq \max\{|X|, \alpha, \aleph_0\}$ and thus $|G| \leq \max\{|X|, \alpha, \aleph_0\}$. $\square$

Proposition 5.11. *The forgetful functor $\mathcal{R}\mathbf{Gr} \rightarrow \mathbf{Set}_f : G \mapsto G$ has a left adjoint $F : \mathbf{Set}_f \rightarrow \mathcal{R}\mathbf{Gr}$.*

Proof. By the adjoint functor theorem (see for example [7, Section 3]) it is sufficient to prove that the category $\mathcal{R}\mathbf{Gr}$ is complete, locally small and the solution set condition is satisfied by the forgetful functor.

Let $(G_i)_{i \in I}$ be a family of $\mathcal{R}$ -groups. Then usual product of groups $\prod_{i \in I} G_i$, with filtration $(\prod_{i \in I} G_i)_k = \prod_{i \in I} (G_i)_k$ and the $\mathcal{R}$ -group structure defined pointwise is an $\mathcal{R}$ -group which clearly is a product in $\mathcal{R}\mathbf{Gr}$. Given two $\mathcal{R}$ -groups G and H and homomorphisms $\phi, \psi : G \rightarrow H$, one easily verifies that $E = \{g \in G \mid \phi(g) = \psi(g)\}$ is an $\mathcal{R}$ -subgroup of G , the injection map $E \rightarrow G$ is the equalizer of ψ and ϕ . The two above facts combined prove that $\mathcal{R}\mathbf{Gr}$ is a complete category. Since the forgetful functor is faithful, $\mathcal{R}\mathbf{Gr}$ is locally small.

We now prove that the forgetful functor satisfies the solution set condition. Let X be a filtered set and consider the class G_X consisting of all pairs (G, f) where G is an $\mathcal{R}$ -group, $f : X \rightarrow G$ a homomorphism of filtered sets and $f(X)$ generates G as an $\mathcal{R}$ -group. Two pairs $(G, f), (H, g) \in G_X$ are said to be isomorphic if there exists an $\mathcal{R}$ -group isomorphism $\phi : G \rightarrow H$ such that $g = \phi f$. Proving that the forgetful functor satisfies the solution set condition amounts to proving that the isomorphism classes of G_X form a set. It follows from Lemma 5.10 that for any $(G, f) \in G_X$, G has cardinality at most $\max\{|X|, \alpha, \aleph_0\}$. As for every set Y of a given cardinality the cardinality of all possible group structures on Y is bounded above by $|Y|^{|Y|^2}$, the cardinality of all possible filtrations on Y is bounded above by $|\mathbb{N}|^{2^{|Y|}}$ and for a given group structure and filtration on Y the number of $\mathcal{R}$ -group structures is bounded above by $|Y|^{|Y|\alpha}$. We conclude that the class G_X is essentially small. $\square$

Let $\mathfrak{g}$ be a Lie $\mathcal{R}$ -algebra. Its *associated graded* Lie $\mathcal{R}$ -algebra is the $\mathcal{R}$ -module $\mathrm{gr} \mathfrak{g} := \bigoplus_{i=1}^{\infty} \mathfrak{g}_i / \mathfrak{g}_{i+1}$, with $[x + \mathfrak{g}_{\deg(x)+1}, y + \mathfrak{g}_{\deg(y)+1}] = [x, y] + \mathfrak{g}_{\deg(x)+\deg(y)+1}$. One easily verifies that this is indeed a Lie $\mathcal{R}$ -algebra and that this construction is functorial.

Similarly, starting from G an $\mathcal{R}$ -group we can construct its *associated graded* Lie $\mathcal{R}$ -algebra. First note that G_i / G_{i+1} becomes an R_i -module by setting $gG_{i+1} + hG_{i+1} = ghG_{i+1}$ and $r(gG_{i+1}) = g^r G_{i+1}$, so it becomes an $\mathcal{R}$ -module as described in Example 2.3. We define $\mathrm{gr} G := \bigoplus_{i=1}^{\infty} G_i / G_{i+1}$, and set $[gG_{i+1}, hG_{j+1}] = ghg^{-1}h^{-1}G_{i+j+1}$. It is a classic result, that $[-, -]$ is a Lie bracket, which is moreover $\mathcal{R}$ -bilinear as for $r \in R_i$ we have

$$\begin{aligned} [rgG_{i+1}, hG_{j+1}] &= g^r hg^{-r} h^{-1} G_{i+j+1} = g^r (hg^{-1}h^{-1})^r G_{i+j+1} \\ &= (ghg^{-1}h^{-1})^r G_{i+j+1} = r[g, h]G_{i+j+1}, \end{aligned}$$

since $(ghg^{-1}h^{-1})^r (hgh^{-1})^r G_{i+j+1} = g^r G_{i+j+1}$ by (G3). Once again it is easily verified that this construction is functorial.

Proposition 5.12. *Let $\mathcal{R}$ be divisible and let H be a complete connected Hopf $\mathcal{R}$ -algebra, then*

$$\phi : \mathrm{gr} \mathbb{P}(H) \rightarrow \mathrm{gr} \mathbb{G}(H) : x + \mathbb{P}(H)_i \mapsto \exp(x) \mathbb{G}(H)_i,$$

is an isomorphism of Lie $\mathcal{R}$ -algebras.

Proof. By earlier observations in Proposition 5.9, the map ϕ is clearly well-defined, bijective and degree preserving. Also for $r \in R_i$ and $y \in \mathbb{P}(H)$ with $\deg(y) = d$

we find $\exp(x)^r \exp(y) + H_{d+1} = rx + y + H_{d+1}$ hence ϕ is a homomorphism of $\mathcal{R}$ -modules, thus an isomorphism because of the earlier observation that it is bijective and degree preserving. At last, it follows from Lemma 5.8 that ϕ is a Lie $\mathcal{R}$ -algebra isomorphism. $\square$

Lemma 5.13. *Let $f : G \rightarrow H$ be a homomorphism of complete $\mathcal{R}$ -groups, then*

- (1) *f is degree preserving if and only if $\text{gr } f$ is injective,*
- (2) *f is a regular surjection if and only if $\text{gr } f$ is surjective.*

Proof. The first claim is easily seen to hold. Also, it is clear that if f is a regular surjection, then $\text{gr } f$ is surjective.

Assume that f is such that $\text{gr } f$ is surjective. We will show that $f(G_i) = H_i$ for any $i \geq 1$. Let $h \in H_i$. As $\text{gr } f$ is surjective, there exists some $g_1 \in G_i$ such that $h^{-1}f(g_1) \in H_{i+1}$. Subsequently, there also exists some $g_2 \in G_{i+1}$ such that $h^{-1}f(g_1)f(g_2) \in H_{i+2}$. We can continue this process in order to obtain $g_j \in G_{i+j}$ such that $h^{-1}f(g_1 \cdots g_j) \in H_{i+j}$. In particular, the limit $g = \prod_{i=1}^{\infty} g_j$ satisfies $h^{-1}f(g)$. At last, note that G_i is closed hence $g \in G_i$. $\square$

We denote the category of connected filtered sets by $\mathbf{Set}_f^{\text{con}}$. We define the functor $\mathcal{R}[-] : \mathbf{Set}_f^{\text{con}} \rightarrow {}_{\mathcal{R}}\mathbf{Hopf}_{pg}^{c,c}$ as $\mathcal{R}[[X]] = \widehat{T(\mathcal{R}X)}$, recall that $\mathcal{R}X$ was defined in Lemma 2.7 and that $\mathcal{R}X$ is connected because X is.

Lemma 5.14. *Let $\mathcal{R}$ be divisible. We have the following adjunctions of functors:*

- (1) *The functor $\mathbb{P}\mathcal{R}[-] : \mathbf{Set}_f^{\text{con}} \rightarrow {}_{\mathcal{R}}\mathbf{Lie}^{c,c}$ is left adjoint to the forgetful functor ${}_{\mathcal{R}}\mathbf{Lie}^{c,c} \rightarrow \mathbf{Set}_f^{\text{con}}$.*
- (2) *The functor $\mathbb{G}\mathcal{R}[-] : \mathbf{Set}_f^{\text{con}} \rightarrow {}_{\mathcal{R}}\mathbf{Gr}^{c,c}$ is left adjoint to the forgetful functor ${}_{\mathcal{R}}\mathbf{Gr}^{c,c} \rightarrow \mathbf{Set}_f^{\text{con}}$.*

Proof. The first claim follows directly from Lemma 2.7, Proposition 2.18 and Theorem 4.7.

Let X be a connected filtered set and let $G = \widehat{F(X)}$ be the completion of the free $\mathcal{R}$ -group generated by X , which is easily seen to be the free complete connected $\mathcal{R}$ -group generated by X . Then there exists a unique homomorphism $f : G \rightarrow \mathbb{G}\mathcal{R}[[X]]$ mapping $x \in X$ to $\exp(x)$. By the first part of the statement we obtain a Lie $\mathcal{R}$ -algebra homomorphism $u : \mathbb{P}\mathcal{R}[[X]] \rightarrow \text{gr } G$ mapping x to $xG_{\deg(x)+1}$, hence also a Lie $\mathcal{R}$ -algebra homomorphism $\text{gr } u : \text{gr } \mathbb{P}\mathcal{R}[[X]] \rightarrow \text{gr } G$ mapping $x + \mathbb{P}(\mathcal{R}[[X]])_{\deg(x)+1}$ to $xG_{\deg(x)+1}$ by the functoriality of gr . Now up to the isomorphism $\text{gr } \mathbb{P}\mathcal{R}[[X]] \cong \text{gr } \mathbb{G}\mathcal{R}[[X]]$ in Proposition 5.12, we find that $\text{gr } u$ and $\text{gr } f$ are mutually inverse hence f is an isomorphism by Lemma 5.13. $\square$

Corollary 5.15. *Let $\mathcal{R}$ be divisible and let X be filtered set, then $\mathbb{P}(\mathcal{R}[[X]])$, respectively $\mathbb{G}(\mathcal{R}[[X]])$, is generated by X as a closed Lie $\mathcal{R}$ -algebra, respectively closed $\mathcal{R}$ -group.*

Proof. This follows directly from the freeness of $\mathbb{P}\mathcal{R}[[X]]$ and $\mathbb{G}\mathcal{R}[[X]]$. $\square$

Corollary 5.16. *Let $\mathcal{R}$ be divisible and let H be a complete connected Hopf $\mathcal{R}$ -algebra. Consider $\{x, y\}$ with filtration $\{x, y\}_1 = \{x, y\}$ and $\{x, y\}_i = \emptyset$ for $i > 1$. There exists some $P(x, y)$ in the free Lie $\mathcal{R}$ -algebra on $\{x, y\}$ such that*

$$\log(\exp(a)\exp(b)) = P(a, b),$$

for all $a, b \in \mathbb{P}(H)$.

Also, there exist $Q(x, y), R(x, y)$ in the free $\mathcal{Q}$ -group on $\{x, y\}$, such that

$$\exp(\log(a) + \log(b)) = Q(a, b), \quad \exp([\log(a), \log(b)]) = R(a, b),$$

for all $a, b \in \mathbb{G}(H)$.

Proof. This follows directly from Corollary 5.15. $\square$

Remark 5.17. It is well-known, see for example [14], that

$$P(x, y) = x + y + \frac{1}{2}[x, y] + \frac{1}{12}([x, [x, y]] + [y, [y, x]]) - \frac{1}{24}[y, [x, [x, y]]] + \dots$$

where further terms involve iterated brackets of order at least 5. This formula is known as the Baker-Campbell-Hausdorff formula. Also

$$Q(g, h) = xy[x, y]^{-\frac{1}{2}}[g, [g, h]]^{\frac{1}{12}}[g, [g, [g, h]]]^{-\frac{1}{24}}[h, [h, [g, h]]]^{\frac{1}{24}} \dots,$$

and

$$R(g, h) = [g, h][g, [g, h]]^{\frac{1}{2}}[h, [g, h]]^{\frac{1}{2}}[g, [g, [g, h]]]^{\frac{1}{3}}[h, [g, [g, h]]]^{\frac{1}{4}}[h, [h, [g, h]]]^{\frac{1}{24}} \dots,$$

where in this case $[a, b] = aba^{-1}b^{-1}$ and further factors involve iterated brackets of order at least 5.

Lemma 5.18. *Let M be a complete $\mathcal{R}$ -module, then $\text{End}_{\mathcal{R}}(M)$ is a complete $\mathcal{R}$ -module.*

Proof. Clearly $\bigcap_{i=0}^{\infty} \text{End}_{\mathcal{R}}(M) = 0$ since $\bigcap_{i=0}^{\infty} M_i = 0$. Given a sequence $f_0, f_1, \dots$ where $f_i \in \text{End}_{\mathcal{R}}(M)_i$, we can define $f \in \text{End}_{\mathcal{R}}(M)$ as $f(m) = \sum_{i=0}^{\infty} f_i(m)$ which is well-defined since M is complete. Then $f = \sum_{i=0}^{\infty} f_i$. $\square$

Let H be a Hopf $\mathcal{R}$ -algebra and $h \in H$. We define $\text{ad}_h : H \rightarrow H : a \mapsto ha - ah$ and for $h \in \mathbb{G}(H)$ we define $\text{conj}_h : H \rightarrow H : a \mapsto hah^{-1}$.

Lemma 5.19. *Let $\mathcal{R}$ be divisible and let H be a complete connected Hopf $\mathcal{R}$ -algebra. Then $\exp(\text{ad}_x) = \text{conj}_{\exp(x)}$ in $\text{End}_{\mathcal{R}}(H)$ for $x \in \mathbb{P}(H)$.*

Proof. One easily verifies that $\phi : H \rightarrow \text{End}_{\mathcal{R}}(H)$ given by $\phi_h(y) = h_{(1)}yS(h_{(2)})$ is a well-defined homomorphism of $\mathcal{R}$ -algebras. For x a primitive element we now find $\exp(\text{ad}_x) = \exp(\phi_x) = \phi_{\exp(x)} = \text{conj}_{\exp(x)}$ which proves the statement. $\square$

Proposition 5.20. *Let $\mathcal{R}$ be divisible and let H be a complete connected Hopf $\mathcal{R}$ -algebra. Then there is a correspondence between closed sub Lie $\mathcal{R}$ -algebras of $\mathbb{P}(H)$ and closed sub $\mathcal{R}$ -groups of $\mathbb{G}(H)$. This correspondence restricts to closed ideals of $\mathbb{P}(H)$ and closed normal sub $\mathcal{R}$ -groups of $\mathbb{G}(H)$.*

Proof. Consider the set $X = \{x, y\}$ with filtration $X_1 = X$, $X_i = \emptyset$ for $i > 1$. For $a, b \in \mathbb{P}(H)$, consider the unique Hopf $\mathcal{R}$ -algebra homomorphism $\mathcal{R}[[X]] \rightarrow H$ mapping $x \mapsto a$ and $y \mapsto b$. By Corollary 5.15 we find that $\exp$ maps the closed sub Lie $\mathcal{R}$ -algebra generated by a, b to the closed sub $\mathcal{R}$ -group generated by $\exp(a), \exp(b)$. This proves the first part of the statement.

The second part now follows directly by Lemma 5.19. $\square$

Let $\mathfrak{g}$ be a complete connected Lie $\mathcal{R}$ -algebra. Then $\mathbb{G}\hat{\mathcal{U}}(\mathfrak{g}) = \exp(\mathfrak{g})$ is a complete $\mathcal{R}$ -group by Proposition 5.9. We transfer the $\mathcal{R}$ -group structure onto $\mathfrak{g}$ through $\exp$ and $\log$, and denote this new group operation by BCH, so $\text{BCH}(g, h) = \log(\exp(g)\exp(h))$ and $g^r = rg$. By Corollary 5.16 we find that this construction is functorial and thus we obtain a functor $\mathbf{Laz} : \mathcal{R}\mathbf{Lie}^{c,c} \rightarrow \mathcal{R}\mathbf{Gr}^{c,c}$.

Theorem 5.21. *Let $\mathcal{R}$ be divisible. The functors $\mathbf{Laz} : \mathcal{R}\mathbf{Lie}^{c,c} \rightarrow \mathcal{R}\mathbf{Gr}^{c,c}$ and $\mathbf{Laz}^{-1} : \mathcal{R}\mathbf{Gr}^{c,c} \rightarrow \mathcal{R}\mathbf{Lie}^{c,c}$ yield an equivalence of categories, where $\mathbf{Laz}(\mathfrak{g}) = (\mathfrak{g}, \cdot)$ with*

$$x \cdot y = \text{BCH}(x, y), rx = x^r \text{ for } x, y \in \mathfrak{g}, r \in R_{\deg(x)},$$

and $\mathbf{Laz}(G) = (G, +, [-, -])$ with

$$x + y = Q(x, y), [x, y] = R(x, y), x^r = rx \text{ for } x, y \in G, r \in R_{\deg(x)}.$$

Proof. From the previous discussion we know that $\mathbf{Laz}$ is a well-defined functor.

If G is complete connected $\mathcal{R}$ -group such that $G = \mathbf{Laz}(\mathfrak{g})$ for some complete connected Lie $\mathcal{R}$ -algebra $\mathfrak{g}$, then by Corollary 5.16 it follows that $\mathbf{Laz}^{-1}(G) = \mathfrak{g}$. We show that every complete connected $\mathcal{R}$ -group G is isomorphic to $\mathbf{Laz}(\mathfrak{g})$ for some complete connected Lie $\mathcal{R}$ -algebra $\mathfrak{g}$. By Lemma 5.14 there exists a regular surjection $\mathbf{Laz}(\mathbb{P}\mathcal{R}[[G]]) \rightarrow G$. Hence $G \cong \mathbf{Laz}(\mathbb{P}\mathcal{R}[[G]])/N$ for some normal sub $\mathcal{R}$ -group N of $\mathbf{Laz}(\mathbb{P}\mathcal{R}[[G]])$. However, by Proposition 5.20 we find that $\mathbf{Laz}(\mathbb{P}\mathcal{R}[[G]])/N = \mathbf{Laz}(\mathbb{P}\mathcal{R}[[G]]/N)$. At last, note that also $\mathbf{Laz}^{-1}$ is functorial. $\square$

6. POST-LIE $\mathcal{R}$ -ALGEBRAS

Lemma 6.1. *Let $(\mathfrak{a}, \triangleright)$ be a post-Lie ring. Then $(\mathfrak{a}, \triangleright)$ is L -nilpotent if and only if $(\mathfrak{a}, \triangleright)$ is left nilpotent and $\mathfrak{a}$ is a nilpotent Lie ring.*

Proof. From the definition of $L^i(\mathfrak{a})$ it is clear that if $(\mathfrak{a}, \triangleright)$ is L -nilpotent then it is left nilpotent and $\mathfrak{a}$ is a nilpotent Lie ring.

Conversely, assume that $(\mathfrak{a}, \triangleright)$ is left nilpotent and $\mathfrak{a}$ is a nilpotent Lie ring. Let $\mathfrak{a} = \mathfrak{a}_1 \supseteq \dots \supseteq \mathfrak{a}_d = 0$ denote the lower central series of $\mathfrak{a}$ and let c be such that $\mathfrak{a}^{c+1} = 1$. If we assume that $L^i(\mathfrak{a}) \subseteq \mathfrak{a}_k$ for some $i, k \geq 1$, then $[L^i(\mathfrak{a}), \mathfrak{a}] \subseteq \mathfrak{a}_{k+1}$ hence $L^{i+1}(\mathfrak{a}) \subseteq (\mathfrak{a} \triangleright L^i(\mathfrak{a})) + \mathfrak{a}_{k+1}$. In particular, $L^{i+c}(\mathfrak{a}) \subseteq \mathfrak{a}^{c+1} + \mathfrak{a}_{k+1} = \mathfrak{a}_{k+1}$. We now conclude by induction that $L^{kc+1}(\mathfrak{a}) = \mathfrak{a}_d = 1$. $\square$

Definition 6.2. A *post-Lie $\mathcal{R}$ -algebra* is a Lie $\mathcal{R}$ -algebra $\mathfrak{a}$ together with a map $\triangleright : \mathfrak{a} \times \mathfrak{a} \rightarrow \mathfrak{a}$ such that

- (P1) $\mathcal{L}_a \in \text{End}_{\mathcal{R}}(\mathfrak{a})$ for all $a \in \mathfrak{a}$, where $\mathcal{L}_a : \mathfrak{a} \rightarrow \mathfrak{a} : b \mapsto a \triangleright b$,
- (P2) $(a + b) \triangleright c = a \triangleright c + b \triangleright c$ for all $a, b, c \in \mathfrak{a}$,
- (P3) $\mathcal{L}_{ra} = r\mathcal{L}_a$ for all $a \in \mathfrak{a}_i$, $r \in R_i$ whenever $\mathcal{L}_a \in \text{End}_{\mathcal{R}}(M)_i$,
- (P4) $\triangleright$ satisfies (1.1) and (1.2).

If $\mathfrak{a}$ is abelian, then $(\mathfrak{a}, \triangleright)$ is a *pre-Lie $\mathcal{R}$ -algebra*.

A homomorphism of post-Lie $\mathcal{R}$ -algebras $f : (\mathfrak{a}, \triangleright) \rightarrow (\mathfrak{b}, \triangleright)$ is a map that respects the Lie $\mathcal{R}$ -algebra structure and moreover $f(a \triangleright b) = f(a) \triangleright f(b)$ for all $a, b \in \mathfrak{a}$.

A sub Lie $\mathcal{R}$ -algebra I of $(\mathfrak{a}, \triangleright)$ is

- (1) a left ideal if $x \triangleright y \in I$ for all $x \in \mathfrak{a}$, $y \in I$,
- (2) a strong left ideal if it is a left ideal and it is an ideal of $\mathfrak{a}$,
- (3) an ideal if it is a strong left ideal and moreover I is an ideal of $\mathfrak{a}^\circ$.

Example 6.3. Let $(\mathfrak{a}, \triangleright)$ be a post-Lie R -algebra. Then $(\mathfrak{a}, \triangleright)$ is a post-Lie $\overline{R}$ -algebra for the filtration $\mathfrak{a}_0 = \mathfrak{a}$, $\mathfrak{a}_i = L^i(\mathfrak{a})$ for $i \geq 1$.

Example 6.4. Let $(\mathfrak{a}, \triangleright)$ be a post-Lie $\mathcal{R}$ -algebra, then

$$\text{Fix}(\mathfrak{a}) := \{a \in \mathfrak{a} \mid b \triangleright a = 0 \text{ for all } b \in \mathfrak{a}\},$$

is a left ideal and

$$\text{Soc}(\mathfrak{a}) := \{a \in \mathfrak{a} \mid a \triangleright b = [a, b] = 0 \text{ for all } b \in \mathfrak{a}\}$$

is an ideal. For the latter statement, note that for $a \in \text{Soc}(\mathfrak{a})$ and $b, c \in \mathfrak{a}$, it follows from (1.1) that $0 = b \triangleright [a, c] = [b \triangleright a, c]$ and from (1.2) that $0 = [a, b] \triangleright c = -(b \triangleright a) \triangleright c$ hence $b \triangleright a \in \text{Soc}(\mathfrak{a})$. As $\{b, a\} = b \triangleright a$, $\text{Soc}(\mathfrak{a})$ is an ideal.

Definition 6.5. Let $\mathfrak{g}$ be a Lie $\mathcal{R}$ -algebra. A map $\delta \in \text{End}_{\mathcal{R}}(\mathfrak{g})$ is a *derivation* if $\delta([x, y]) = [\delta(x), y] + [x, \delta(y)]$ for all $x, y \in \mathfrak{g}$. We define $\text{Der}(\mathfrak{g})$ as the set of all derivations of $\mathfrak{g}$, which are easily seen to form a sub Lie $\mathcal{R}$ -algebra of $\text{End}_{\mathcal{R}}(\mathfrak{g})$ under the usual commutator Lie bracket.

Lemma 6.6. *Let $\mathfrak{g}$ be a Lie $\mathcal{R}$ -algebra, then the direct product of $\mathcal{R}$ -modules $\text{aff}(\mathfrak{g}) := \mathfrak{g} \oplus \text{Der}(\mathfrak{g})$ together with Lie bracket*

$$[(g, \delta), (h, \chi)] = ([g, h] + \delta(h) - \chi(g), [\delta, \chi]),$$

is a Lie $\mathcal{R}$ -algebra. We denote the projection onto $\mathfrak{g}$ by $\text{pr}_{\mathfrak{g}}$.

Definition 6.7. Let $\mathfrak{a}$ be a Lie $\mathcal{R}$ -algebra, a sub Lie $\mathcal{R}$ -algebra $\mathfrak{g}$ of $\text{aff}(\mathfrak{a})$ is *t-surjective*, respectively *t-bijective* if $\text{pr}_{\mathfrak{a}} : \mathfrak{g} \rightarrow \mathfrak{a}$ is surjective, respectively bijective.

Proposition 6.8. *Let $\mathfrak{a}$ be a Lie $\mathcal{R}$ -algebra. Then there is a bijective correspondence between maps $\triangleright : \mathfrak{a} \times \mathfrak{a} \rightarrow \mathfrak{a}$ making $(\mathfrak{a}, \triangleright)$ into a post-Lie $\mathcal{R}$ -algebra and t-bijective sub Lie $\mathcal{R}$ -algebras of $\text{aff}(\mathfrak{a})$.*

Proof. Let $\mathfrak{a}$ be a Lie $\mathcal{R}$ -algebra and $\triangleright : \mathfrak{a} \times \mathfrak{a} \rightarrow \mathfrak{a}$ any map. The maps $\mathcal{L}_a : \mathfrak{a} \rightarrow \mathfrak{a} : b \mapsto a \triangleright b$, for $a \in \mathfrak{a}$, are contained in $\text{End}_{\mathcal{R}}(\mathfrak{a})$ if and only if (P1) holds. The $\mathcal{L}_a$ are derivations of $\mathfrak{a}$ if and only if (1.1) holds. Suppose that all of the previous assumptions are satisfied and consider $\mathfrak{g} = \{(x, \mathcal{L}_x) \mid x \in \mathfrak{a}\} \in \text{aff}(\mathfrak{a})$. Conditions (P2) and (P3) are satisfied precisely if $\mathfrak{g}$ is a sub $\mathcal{R}$ -module of $\text{aff}(\mathfrak{a})$. We see that $\mathfrak{g}$ is a sub Lie $\mathcal{R}$ -algebra if $\mathcal{L}_{[x, y] + x \triangleright y - y \triangleright x} = [\mathcal{L}_x, \mathcal{L}_y]$ which is equivalent to (1.2). This proves the correspondence. $\square$

Corollary 6.9. *Let $(\mathfrak{a}, \triangleright)$ be a post-Lie $\mathcal{R}$ -algebra. We obtain a new Lie $\mathcal{R}$ -algebra structure $\mathfrak{a}^\circ$ on $\mathfrak{a}$ with filtration $\mathfrak{a}_i^\circ = \{a \in \mathfrak{a}_i \mid \mathcal{L}_a \in \text{Der}(\mathfrak{a})_i\}$ and Lie bracket*

$$\{a, b\} := [a, b] + a \triangleright b - b \triangleright a.$$

Remark 6.10. Note that the at first somewhat artificial definition of a post-Lie $\mathcal{R}$ -algebra becomes more natural when we remark that (P1) – (P3) state precisely that $\mathfrak{a}^\circ \times \mathfrak{a} \rightarrow \mathfrak{a}$ is an $\mathcal{R}$ -bilinear map.

Proof. Let $\mathfrak{g}$ be the associated t-bijective sub Lie $\mathcal{R}$ -algebra of $\text{aff}(\mathfrak{a})$ as in Proposition 6.8. The statement follows by transferring the Lie $\mathcal{R}$ -algebra structure of $\mathfrak{g}$ onto $\mathfrak{a}$ through the bijection $\text{pr}_{\mathfrak{a}} : \mathfrak{g} \rightarrow \mathfrak{a}$. $\square$

Definition 6.11. Let $(\mathfrak{a}, \triangleright)$ be a post-Lie $\mathcal{R}$ -algebra, then $(\mathfrak{a}, \triangleright)$ is

- (1) *connected* if both $\mathfrak{a}$ and $\mathfrak{a}^\circ$ are connected Lie $\mathcal{R}$ -algebras,
- (2) *complete* if both $\mathfrak{a}$ and $\mathfrak{a}^\circ$ are complete Lie $\mathcal{R}$ -algebras.

The category of complete connected post-Lie $\mathcal{R}$ -algebras is denoted $\mathcal{R}\text{PLA}^{c,c}$.

Example 6.12. Let $(\mathfrak{a}, \triangleright)$ be a post-Lie $\mathcal{R}$ -algebra, then the associated post-Lie $\overline{\mathcal{R}}$ -algebra as in Example 6.3 is connected. If $(\mathfrak{a}, \triangleright)$ is L -nilpotent, it is complete.

Let $\mathbb{F}$ be a field of characteristic 0 and $(\mathfrak{a}, \triangleright)$ be a finite dimensional pre-Lie $\mathbb{F}$ -algebra. Then the classical definition of completeness, as coined in [19], is that for every $a \in \mathfrak{a}$, the map $r_a : \mathfrak{a} \rightarrow \mathfrak{a} : b \mapsto b + a \triangleright b$ is a bijection. However, if $\mathfrak{a}^\circ$ is nilpotent, then [32, Proposition 2] states that all maps r_a are bijective if and only if left multiplication maps $\mathcal{L}_a$ are nilpotent. By Engel's theorem, this is equivalent with the existence of some n such that $\mathcal{L}_{a_1} \cdots \mathcal{L}_{a_n} = 0$ for all $a_1, \dots, a_n \in \mathfrak{a}$. It follows that this notion of completeness coincides with what we call left nilpotency. Therefore, a pre-Lie $\mathbb{F}$ -algebra $(\mathfrak{a}, \triangleright)$ with $\mathfrak{a}^\circ$ nilpotent is complete if and only if its associated pre-Lie $\overline{\mathcal{F}}$ -algebra (as in Example 6.3) is complete. Note that the condition that all $\mathcal{L}_a$ are nilpotent also appears in [10, Theorem 3.1].

Theorem 6.13. *Let $(\mathfrak{a}, \triangleright)$ be an L -nilpotent post-Lie ring of class k . Then $\mathfrak{a}^\circ$ is nilpotent of class at most k and the k -th term in the lower central series of $\mathfrak{a}^\circ$ is contained in $\text{Ann}(\mathfrak{a})$, where $\text{Ann}(\mathfrak{a}) := \text{Soc}(\mathfrak{a}) \cap \text{Fix}(\mathfrak{a})$.*

Proof. Let $(\mathfrak{a}, \triangleright)$ be L -nilpotent of class k and consider it as a post-Lie $\overline{\mathbb{Z}}$ -algebra. Then $\mathfrak{a}_{k+1}^\circ \subseteq \mathfrak{a}_{k+1} = 0$, hence in particular $\mathfrak{a}^\circ$ is nilpotent of class at most k . Also, since $\mathfrak{a}_k^\circ = \mathfrak{a}_k \cap \{a \in \mathfrak{a} \mid \mathcal{L}_a \in \text{End}_{\mathcal{R}}(\mathfrak{a})_k\}$ but $\text{End}_{\mathcal{R}}(\mathfrak{a})_k = 0$ and clearly $\mathfrak{a}_k \in Z(\mathfrak{a}) \cap \text{Fix}(\mathfrak{a})$, the statement follows. $\square$

7. SKEW $\mathcal{R}$ -BRACES

We start by noting that one can prove Lemma 1.2 using almost the same proof as for Lemma 6.1.

Definition 7.1. A skew $\mathcal{R}$ -brace is an $\mathcal{R}$ -group $(A, \cdot)$ with an additional group structure $(A, \circ)$ such that

- (S1) $a \circ (b \cdot c) = (a \circ b) \cdot a^{-1} \cdot (a \circ c)$,
- (S2) $a * b \in A_{i+1}$ and $\lambda_a(b^r) = \lambda_a(b)^r$ for $a \in A$, $b \in A_i$, $r \in R_i$,

where we define $a * b = a^{-1} \cdot (a \circ b) \cdot b^{-1}$ and $\lambda_a(b) = a^{-1} \cdot (a \circ b)$. An $\mathcal{R}$ -brace is an skew $\mathcal{R}$ -brace with an abelian additive group. A homomorphism of skew $\mathcal{R}$ -braces $f : (A, \cdot, \circ) \rightarrow (B, \cdot, \circ)$ is a map such that $f : (A, \cdot) \rightarrow (B, \cdot)$ is a homomorphism of $\mathcal{R}$ -groups and $f : (A, \circ) \rightarrow (B, \circ)$ is a group homomorphism.

A sub $\mathcal{R}$ -group I of $(A, \cdot)$ is

- (1) a *sub skew $\mathcal{R}$ -brace* if I is a subgroup of $(A, \circ)$,
- (2) a *left ideal* if $\lambda_a(b) \in I$ for all $a \in A$, $b \in I$,
- (3) a *strong left ideal* if I is a left ideal and also a normal subgroup of $(A, \cdot)$,
- (4) an *ideal* if I is a strong left ideal and also a normal subgroup of $(A, \circ)$.

Example 7.2. Let $(A, \cdot, \circ)$ be a skew brace, then the filtration $A_0 = A$, $A_i = L^i(A)$ for $i \geq 1$ makes it into a skew $\overline{\mathbb{Z}}$ -brace.

Example 7.3. Let $(A, \cdot, \circ)$ be an R -brace, see [16, Definition 2], and consider the associated $\mathbb{Z}$ -brace as in Example 7.2. Then $L^i(A)$ is a sub R -modules of A for every $i \geq 1$, since $a * (rb) = r(a * b)$ for $r \in R$, $a, b \in A$. In this way the obtained $\overline{\mathbb{Z}}$ -brace has a natural $\overline{R}$ -brace structure. Conversely, any $\overline{R}$ -brace becomes an R -brace if we simply forget the filtration.

Definition 7.4. For G an $\mathcal{R}$ -group, we define $\text{Aut}(G)$ as all $\mathcal{R}$ -group automorphisms of G . We define a filtration $\text{Aut}(G)_i$ by

$$\text{Aut}(G)_i = \{f \mid f(g)g^{-1} \in G_{i+j}, \text{ for all } g \in G_j\}$$

Remark 7.5. In Definition 7.1 it follows from (S1) that $\lambda_a(b \cdot c) = \lambda_a(b) \cdot \lambda_a(c)$, hence (S2) can be rephrased as $\lambda_a \in \text{Aut}(A, \cdot)_1$.

Lemma 7.6. Let G be an $\mathcal{R}$ -group, then $\text{Hol}(G) := G \rtimes \text{Aut}(G)$ with $\text{Hol}(G)_i = G_i \rtimes \text{Aut}(G)_i$ is a filtered group. We denote the projection onto G by pr_G .

Definition 7.7. Let G be an $\mathcal{R}$ -group. A subgroup $H \leq \text{Hol}(G)$ is *transitive*, respectively *regular*, if $\text{pr}_G : H \rightarrow G$ is surjective, respectively bijective.

Remark 7.8. Note that if we consider the natural action of $\text{Hol}(G)$ on G , i.e. $(g, \lambda) \cdot h = g\lambda(h)$, then transitive subgroups of $\text{Hol}(G)$ are precisely the ones such that the associated action is transitive.

Proposition 7.9. Let $(A, \cdot)$ be an $\mathcal{R}$ -group. There is a bijective correspondence between group structures $(A, \circ)$ making $(A, \cdot, \circ)$ into a skew $\mathcal{R}$ -brace and regular subgroups of $\text{Hol}(A, \cdot)$.

Proof. We only sketch the proof, as this is a slight variation of [18, Theorem 4.2]. Let $(A, \circ)$ be a group operation such that $(A, \cdot, \circ)$ is a skew $\mathcal{R}$ -brace. Define $G = \{(a, \lambda_a) \mid a \in A\} \leq \text{Hol}(A)$, then G is a regular subgroup of $\text{Hol}(A)$. Conversely,

if G is a regular subgroup of $\text{Hol}(A)$, we transfer the group structure of G onto A with the help of the bijection $\text{pr}_A : G \rightarrow A$ and we denote this new group operation by $\circ$. Then $(A, \cdot, \circ)$ becomes a skew $\mathcal{R}$ -brace. $\square$

Corollary 7.10. *Let $(A, \cdot, \circ)$ be a skew $\mathcal{R}$ -brace. Then $(A, \circ)$ with the filtration $(A, \circ)_i = A_i \cap \{a \in A \mid \lambda_a \in \text{Aut}(A)_i\}$ is a filtered group.*

Proof. Let G be the associated regular subgroup of $\text{Hom}(A, \cdot)$ as in Proposition 7.9 with filtration coming from $\text{Hom}(A, \cdot)$. Transferring the filtration of G onto $(A, \circ)$ through the isomorphism $(A, \circ) \rightarrow G : a \mapsto (a, \lambda_a)$ now yields the statement. $\square$

Definition 7.11. Let $(A, \cdot, \circ)$ be a skew $\mathcal{R}$ -brace, then $(A, \cdot, \circ)$ is

- (1) *connected* if both $(A, \cdot)$ and $(A, \circ)$ are connected filtered groups,
- (2) *complete* if both $(A, \cdot)$ and $(A, \circ)$ are complete filtered groups.

Example 7.12. Let $(A, \cdot, \circ)$ be a skew brace, then the associated skew $\overline{\mathbb{Z}}$ -brace as in Example 7.2 is connected. If $(A, \cdot, \circ)$ is L -nilpotent, it is complete. The same applies for the $\overline{R}$ -braces as obtained in Example 7.3.

The proof of Theorem 1.3 similar to the proof of Theorem 6.13.

8. A LAZARD CORRESPONDENCE FOR POST-LIE $\mathcal{R}$ -ALGEBRAS AND SKEW $\mathcal{R}$ -BRACES

Definition 8.1. For $\mathfrak{g}$ a Lie $\mathcal{R}$ -algebra we define $\text{Aut}(\mathfrak{g})$ as the set of all Lie $\mathcal{R}$ -algebra automorphisms with filtration $\text{Aut}(\mathfrak{g})_i = \text{Aut}(\mathfrak{g}) \cap (\text{id} + \text{End}_{\mathcal{R}}(\mathfrak{g})_i)$.

Definition 8.2. For a filtered set X we define X^+ as the connected filtered subset consisting of all elements of X_1 . The same notation is used also for $\mathcal{R}$ -modules and $\mathcal{R}$ -groups.

Lemma 8.3. *Let $\mathcal{R}$ be divisible and let $\mathfrak{g}$ be a complete Lie $\mathcal{R}$ -algebra. Then $\text{Aut}(\mathfrak{g})_i = \exp(\text{Der}(\mathfrak{g})_i)$ for $i \geq 1$. In particular, the group $\text{Aut}(\mathfrak{g})^+$ has a canonical $\mathcal{R}$ -group structure such that $\exp : \text{Laz}(\text{Der}(\mathfrak{g})^+) \rightarrow \text{Aut}(\mathfrak{g})^+$ is an $\mathcal{R}$ -group isomorphism.*

Proof. Let $X = \{x\}$ with the filtration $X_1 = X$, $X_2 = \emptyset$ and let $\mathcal{R}[[x]] := \mathcal{R}[[X]]$. For any $\mathcal{R}$ -module homomorphism $\phi : \mathcal{R}[[x]] \rightarrow \text{End}_{\mathcal{R}}(\mathfrak{g})$ the set

$$S = \{h \in \mathcal{R}[[x]] \mid [-, -](\phi_{h(1)} \otimes \phi_{h(2)}) = \phi_h[-, -]\}$$

is a closed sub $\mathcal{R}$ -algebra of $\mathcal{R}[[x]]$. As $\{x\}$ and $\{\exp(x)\}$ generate $\mathcal{R}[[x]]$ as a closed sub $\mathcal{R}$ -algebra, we find that $\phi_x \in \text{Der}(\mathfrak{g})_i$ if and only if $\phi_{\exp(x)} \in \text{Aut}(\mathfrak{g})_i$. For any element f of $\text{End}_{\mathcal{R}}(\mathfrak{g})_1$ there exists a unique $\mathcal{R}$ -algebra homomorphism $\mathcal{R}[[x]] \rightarrow \text{End}_{\mathcal{R}}(\mathfrak{g})$ mapping x to f , so for every element $g \in 1 + \text{End}_{\mathcal{R}}(\mathfrak{g})_1$ there exists a unique $\mathcal{R}$ -algebra homomorphism $\mathcal{R}[[x]] \rightarrow \text{End}_{\mathcal{R}}(\mathfrak{g})$ mapping $\exp(x)$ to g . This implies for any $i \geq 1$ that $\delta \in \text{Der}(\mathfrak{g})_i$ if and only if $\exp(\delta) \in \text{Aut}(\mathfrak{g})_i$. $\square$

Lemma 8.4. *Let $\mathcal{R}$ be divisible and let $\mathfrak{g}$ be a complete Lie $\mathcal{R}$ -algebra. Then $\text{Aut}(\hat{\mathcal{U}}(\mathfrak{g}))_i = \exp(\text{Der}(\hat{\mathcal{U}}(\mathfrak{g}))_i)$ for $i \geq 1$. In particular, the group $\text{Aut}(\hat{\mathcal{U}}(\mathfrak{g}))^+$ has a canonical $\mathcal{R}$ -group structure such that $\exp : \text{Laz}(\text{Der}(\hat{\mathcal{U}}(\mathfrak{g}))^+) \rightarrow \text{Aut}(\hat{\mathcal{U}}(\mathfrak{g}))^+$ is an $\mathcal{R}$ -group isomorphism.*

Proof. The proof is analogous to the proof of Lemma 8.3. $\square$

Lemma 8.5. *Let $\mathcal{R}$ be divisible and let $\mathfrak{g}$ be a complete connected Lie $\mathcal{R}$ -algebra, then $\text{Aut}(\mathfrak{g}) = \text{Aut}(\text{Laz}(\mathfrak{g}))$.*

Proof. We know by Theorem 5.21 that automorphisms of $\mathfrak{g}$ and G coincide. It remains to verify that the filtrations coincide. Denote $H = \hat{\mathcal{U}}(\mathfrak{g})$, let $x, y \in \mathbb{P}(H)$ and denote $d = \deg(x + y)$. Then $[x, y] + H_{d+1} = [x, y] - [x + y, y] + H_{d+1} = H_{d+1}$, hence $x + y + H_{d+1} = \text{BCH}(x, y) + H_{d+1}$ by Remark 5.17. In particular, for $\delta \in \text{Der}(\mathfrak{g})^+$ we find $\deg(\text{BCH}(\exp(\delta)(x), -x)) = \deg(\exp(\delta)(x) - x)$, thus the degree of $\exp(\delta)$ in $\text{Aut}(\mathbf{Laz}(\mathfrak{g}))$ equals the degree of $\exp(\delta) - \text{id}$ in $\text{End}_{\mathcal{R}}(\mathfrak{g})$, which is easily seen to equal the degree of δ in $\text{End}_{\mathcal{R}}(\mathfrak{g})$ (or equivalently in $\text{Der}(\mathfrak{g})^+$). $\square$

Proposition 8.6. *Let $\mathcal{R}$ be divisible and let $\mathfrak{a}$ be a complete connected Lie $\mathcal{R}$ -algebra, then*

$$\gamma : \mathbf{Laz}(\mathfrak{aff}(\mathfrak{a})^+) \rightarrow \text{Hol}(\mathbf{Laz}(\mathfrak{a}))^+ : (a, \delta) \mapsto (V(a, \delta), \exp(\delta)),$$

with $V(a, \delta) = \text{pr}_{\mathfrak{a}}(\text{BCH}((a, \delta), (0, -\delta)))$ is an isomorphism of filtered groups.

In particular, $\text{Hol}(\mathbf{Laz}(\mathfrak{a}))^+$ has an $\mathcal{R}$ -group structure such that f is an isomorphism of $\mathcal{R}$ -groups.

Proof. We will work inside of $H = \hat{\mathcal{U}}(\mathfrak{aff}(\mathfrak{a})^+)$ and we identify $\mathfrak{a}$ and $\text{Der}(\mathfrak{a})^+$ with their corresponding sub Lie $\mathcal{R}$ -algebras of $\mathfrak{aff}(\mathfrak{a})^+ = \mathbb{P}(H)$. Note that by Lemma 5.19 and Lemma 8.3 the map

$$f : \exp(\mathfrak{a}) \exp(\text{Der}(\mathfrak{a})^+) \rightarrow \text{Hol}(\mathbf{Laz}(\mathfrak{a}))^+ : \exp(a, 0) \exp(0, \delta) \mapsto (a, \exp(\delta))$$

is a filtered group isomorphism.

Let $(a, \delta) \in \mathfrak{aff}(\mathfrak{a})^+$, then $\log(\exp(a, \delta) \exp(0, -\delta)) + \mathfrak{aff}(\mathfrak{a})^+ = \mathfrak{aff}(\mathfrak{a})^+$, hence $\exp(a, \delta) \exp(0, -\delta) \in \exp(\mathfrak{a})$. We conclude that

$$\exp(a, \delta) = (\exp(a, \delta) \exp(0, -\delta)) \exp(0, \delta) \in \exp(\mathfrak{a}) \exp(\text{Der}(\mathfrak{a})^+),$$

meaning that

$$\mathbf{Laz}(\mathfrak{aff}(\mathfrak{a})^+) \rightarrow \exp(\mathfrak{a}) \exp(\text{Der}(\mathfrak{a})^+) : (a, \delta) \mapsto (\exp(a, \delta) \exp(0, -\delta)) \exp(0, \delta),$$

is an $\mathcal{R}$ -group isomorphism. Composing this isomorphism with f we find indeed that γ is an isomorphism of groups. $\square$

Remark 8.7. We can explicitly write $\gamma^{-1}(a, \lambda) = (U(a, \lambda), \log(\lambda))$, where $U(a, \lambda) = \text{pr}_{\mathfrak{a}}(P((a, \lambda), (0, \lambda^{-1})))$ with P as in Corollary 5.16. The proof for this is analogous to the one of Proposition 8.6.

Definition 8.8. If $\mathcal{R}$ is a divisible sequence of rings then a skew $\mathcal{R}$ -brace $(A, \cdot, \circ)$ is *strict* if the corresponding regular subgroup of $\text{Hol}(A, \cdot)$ is a sub $\mathcal{R}$ -group, with the $\mathcal{R}$ -group structure of $\text{Hol}(A, \cdot)$ as in Proposition 8.6.

The following example shows that there exist $\mathcal{R}$ -braces that are not strict.

Example 8.9. Let R be a commutative ring of characteristic different from 2 and $f : (R, +) \rightarrow (R, +)$ a group endomorphism. Then the sequence $\mathcal{R} = R \rightarrow R \rightarrow R \rightarrow 0 \rightarrow \dots$ is divisible. On $M = R \oplus R$ we consider the filtration $M_0 = M_1 = M$, $M_2 = R \oplus 0$, $M_i = 0$ for $i \geq 3$, with the canonical $\mathcal{R}$ -module structure coming from the R -module structure. On M we define

$$(x_1, y_1) \circ (x_2, y_2) = (x_1 + x_2 + f(y_1)y_2, y_1 + y_2),$$

which is easily verified to be a group structure. Also

$$\begin{aligned} (x_1, y_1) \circ ((x_2, y_2) + (x_3, y_3)) &= (x_1 + x_2 + x_3 + f(y_1)(y_2 + y_3), y_1 + y_2 + y_3) \\ &= (x_1 + x_2 + f(y_1)y_2, y_1 + y_2) - (x_1, y_1) + (x_1 + x_3 + f(y_1)y_3, y_1 + y_3) \\ &= (x_1, y_1) \circ (x_2, y_2) - (x_1, y_1) + (x_1, y_1) \circ (x_3, y_3), \end{aligned}$$

and $\lambda_{(x_1, y_1)}(x_2, y_2) = (x_2 + f(y_1)y_2, y_2)$, hence $\lambda_{(x, y)} \in \text{Aut}^+(M)$ meaning that $(M, +, \circ)$ is an $\mathcal{R}$ -brace. We find $\log(\lambda_{(a, b)})(x, y) = (f(b)y, 0)$ and thus $\lambda_{(a, b)}^r$ is

given by $(x, y) \mapsto (x + rf(b)y, y)$. Hence, $\lambda_{(a,b)}^r = \lambda_{(c,d)}$ if and only if $rf(b) = f(d)$, hence $\lambda(M)$ is a sub $\mathcal{R}$ -group of $\text{Aut}(M)^+$ if and only if $f(R)$ is a left ideal of R . Note that $\lambda(M)$ being a sub $\mathcal{R}$ -group of $\text{Aut}(M)^+$ is a necessary condition for $(M, +, \circ)$ to be strict.

Proposition 8.10. *Let $\mathcal{R}$ be divisible, let $\mathfrak{a}$ be a complete connected Lie $\mathcal{R}$ -algebra and set $A = \mathbf{Laz}(\mathfrak{a})$. Then γ yields a bijective correspondence between closed sub Lie $\mathcal{R}$ -algebras $\mathfrak{g}$ of $\mathfrak{aff}(\mathfrak{a})^+$ and closed sub $\mathcal{R}$ -groups G of $\text{Hol}(A)^+$. Moreover, $\text{pr}_{\mathfrak{a}} : \mathfrak{g} \rightarrow \mathfrak{a}$ is injective, respectively surjective, if and only if $\text{pr}_A : \gamma(\mathfrak{g}) \rightarrow A$ is injective, respectively surjective.*

Proof. The first part follows directly from Proposition 5.20 and Proposition 8.6.

Assume that $\mathfrak{g}$ is a closed Lie $\mathcal{R}$ -algebra of $\mathfrak{aff}(\mathfrak{a})$ such that $\text{pr}_{\mathfrak{a}} : \mathfrak{g} \rightarrow \mathfrak{a}$ is not injective. As this map is $\mathcal{R}$ -linear, this means that there exists some non-zero element $\delta \in \mathfrak{g} \cap \text{Der}^+(\mathfrak{a})$. But then clearly $V(\delta) = \text{BCH}(\delta, -\delta) = 0 = V(0)$, meaning that V is not injective on $\mathfrak{g}$ and thus $\text{pr}_A : \gamma(\mathfrak{g}) \rightarrow A$ is not injective.

Conversely, assume that there exist $(a, \lambda), (a, \lambda') \in \gamma(\mathfrak{g})$ with $\lambda \neq \lambda'$. Then $(a, \lambda)^{-1}(a, \lambda') = (0, \lambda^{-1}\lambda') \in f(\mathfrak{g})$. Hence there exists some $(b, \delta) \in \mathfrak{aff}(\mathfrak{a})$ with $\delta \neq 0$ such that $V(b, \delta) = 0$ hence $b = 0$.

Assume that $\text{pr}_{\mathfrak{a}} : \mathfrak{g} \rightarrow \mathfrak{a}$ is surjective. We claim that $\text{pr}_A \gamma = V : \mathfrak{g} \rightarrow \mathfrak{a}$ is surjective. Since pr_A and γ are continuous, the set $V(\mathfrak{g})$ is closed in $\mathfrak{a}$. Assume that there exists some $a \in A \setminus \text{pr}_A \gamma(\mathfrak{g})$, then there exists a d such that $V(\mathfrak{g}) \cap (a + \mathfrak{a}_d) \neq \emptyset$ but $V(\mathfrak{g}) \cap (a + \mathfrak{a}_{d+1}) = \emptyset$. Let $(b, \delta) \in \mathfrak{g}$ such that $V(b, \delta) \in a + \mathfrak{a}_d$ and let $(b', \delta') \in \mathfrak{g}$ such that $b' = \text{BCH}(-V(b, \delta), a)$. Since $b' \in \mathfrak{a}_d$, we have

$$(8.1) \quad b' + \mathfrak{a}_{d+1} = V(b', \delta') + \mathfrak{a}_{d+1} = \exp(\delta)(V(b', \delta')) + \mathfrak{a}_{d+1}.$$

Hence

$$\begin{aligned} V(\text{BCH}((b, \delta), (b', \delta'))) + \mathfrak{a}_{d+1} &= \text{BCH}(V(b, \delta), \exp(\delta)V(b', \delta')) + \mathfrak{a}_{d+1} \\ &= \text{BCH}(V(b, \delta), \text{BCH}(-V(b, \delta), a)) + \mathfrak{a}_{d+1} \\ &= a + \mathfrak{a}_{d+1}, \end{aligned}$$

where the first equality follows from the fact that γ is a group homomorphism. We find $\text{BCH}(-V(\text{BCH}((b, \delta), (b', \delta'))), a) \in a + \mathfrak{a}_{d+1}$, which is a contradiction.

Conversely, assume that $V(\mathfrak{g}) = \mathfrak{a}$. Clearly $\text{pr}_{\mathfrak{a}}(\mathfrak{g})$ is closed in $\mathfrak{a}$ because $\text{pr}_{\mathfrak{a}}$ is continuous. Assume that there exists some $a \in \mathfrak{a} \setminus \text{pr}_{\mathfrak{a}}(\mathfrak{g})$ and let d be such that $\text{pr}_{\mathfrak{a}}(\mathfrak{g}) \cap (a + \mathfrak{a}_d) \neq \emptyset$ but $\text{pr}_{\mathfrak{a}}(\mathfrak{g}) \cap (a + \mathfrak{a}_{d+1}) = \emptyset$. Choose $(b, \delta) \in \mathfrak{g}$ such that $b \in \mathfrak{a}_d$ and choose $(b', \delta') \in \mathfrak{g}$ such that $V(b', \delta') = b - a$. Using (8.1) we find

$$\text{pr}_{\mathfrak{a}}(h + h') + \mathfrak{a}_{d+1} = b + V(h') + \mathfrak{a}_{d+1} = a + \mathfrak{a}_{d+1},$$

hence $\text{pr}_{\mathfrak{a}}(h + h') \in a + \mathfrak{a}_{d+1}$, which is a contradiction. $\square$

Remark 8.11. Transitive subgroups of the holomorph of a finite group play an essential role in the theory of Hopf-Galois structures, see for example [13, 38, 23]. Although in general $\text{Hol}^+(A)$ is a strict subgroup of $\text{Hol}(A)$ one can prove that if A is a p -group then there exists a finite filtration of A such that A is connected and $\text{Hol}(A)^+$ is a Sylow p -subgroup of $\text{Hol}(A)$. If $|A| = p^k$ for $k < p$, then such a filtration has length at most p so we can see A as a $\mathbb{Z}/p^k\mathbb{Z}$ -group. Hence in such a case, we can classify all regular p -groups of $\text{Hol}(A)$ by classifying the t -surjective sub Lie rings of $\mathfrak{aff}(\mathbf{Laz}^{-1}(A))$.

Definition 8.12. Let $\mathcal{R}$ be divisible and let $(A, \cdot, \circ)$ be a complete connected skew $\mathcal{R}$ -brace. Then $(A, \cdot, \circ)$ is *strict* if it associated regular subgroup G of $\text{Hol}(A, \cdot)^+$ is a sub $\mathcal{R}$ -group, with the $\mathcal{R}$ -group structure as in Proposition 8.6, in this case we transfer the $\mathcal{R}$ -group structure of G onto $(A, \circ)$ through their canonical bijection. A

sub skew $\mathcal{R}$ -brace of $(A, \cdot, \circ)$ is *strict* if it is a sub $\mathcal{R}$ -group of $(A, \circ)$. The category of strict complete connected skew $\mathcal{R}$ -braces is denoted $\mathcal{R}\mathbf{SB}^{c,c,s}$.

Lemma 8.13. *Let $\mathcal{R}$ be divisible and let $(A, \cdot, \circ)$ be a strict complete connected skew $\mathcal{R}$ -brace. Then every left ideal is a strict sub skew $\mathcal{R}$ -brace.*

Proof. Let I be a left ideal of $(A, \cdot, \circ)$. Since $a \circ b = a \cdot \lambda_a(b)$, it is always true that a left ideal is a sub skew $\mathcal{R}$ -brace. It remains to show that it is strict, which is equivalent to $I' = \{(x, \lambda_x) \mid x \in I\}$ being a sub $\mathcal{R}$ -group of $\text{Hol}(A, \cdot)^+$. However, $I' = G \cap (I \rtimes \text{Aut}(A, \cdot)^+)$ where G is the associated regular subgroup of $\text{Hol}(A, \cdot)^+$. By assumption, G is a sub $\mathcal{R}$ -group. It is easily verified that $\gamma^{-1}(I \rtimes \text{Aut}(A, \cdot)^+) = I \oplus \text{Der}(\mathfrak{a})^+ \leq \text{aff}(\mathfrak{a})^+$ with $\mathfrak{a} = \mathbf{Laz}(A, \cdot)$. Since $I \oplus \text{Der}(\mathfrak{a})^+$ is a closed sub $\mathcal{R}$ -algebra of $\text{aff}(\mathfrak{a})^+$, we find that $I \rtimes \text{Aut}(A, \cdot)^+$ is a sub $\mathcal{R}$ -group of $\text{Hol}(A, \cdot)$. We conclude that I' is an intersection of two sub $\mathcal{R}$ -groups, hence it is a sub $\mathcal{R}$ -group itself. $\square$

Proposition 8.14. *Let $\mathcal{R}$ be divisible and let $(\mathfrak{a}, \triangleright)$ be a complete connected post-Lie $\mathcal{R}$ -algebra. Then*

$$W : \mathfrak{a} \rightarrow \mathfrak{a} : a \mapsto V(a, \mathcal{L}_a),$$

is a bijective map and $\mathbf{S}(\mathfrak{a}, \triangleright) = (\mathfrak{a}, \cdot, \circ)$ with $(\mathfrak{a}, \cdot) = \mathbf{Laz}(\mathfrak{a})$, $a \circ b = a \cdot \exp(\mathcal{L}_{\Omega(a)})(b)$ is a strict complete connected skew $\mathcal{R}$ -brace, where $\Omega := W^{-1}$. Moreover, $W : \mathbf{Laz}(\mathfrak{a}^\circ) \rightarrow (\mathfrak{a}, \circ)$ is an $\mathcal{R}$ -group isomorphism.

Proof. Let $(\mathfrak{a}, \triangleright)$ be a complete connected post-Lie $\mathcal{R}$ -algebra and let $\mathfrak{g}$ be the regular sub Lie $\mathcal{R}$ -algebra of $\text{aff}(\mathfrak{a})^+$ associated to $(\mathfrak{a}, \triangleright)$. Then W is a bijection since it is the composition of the bijections $\mathfrak{a} \rightarrow \mathfrak{g} : a \mapsto (a, \mathcal{L}_a)$ and $V : \mathfrak{g} \rightarrow \mathfrak{a}$, recall that the latter is a bijection by Proposition 5.20.

By Proposition 5.20 we find that $\gamma(\mathfrak{g})$ is a closed regular sub $\mathcal{R}$ -group of $\text{Hol}(\mathbf{Laz}(\mathfrak{a}))^+$, thus we obtain a strict complete connected skew $\mathcal{R}$ -brace $(\mathfrak{a}, \cdot, \circ)$ where $(\mathfrak{a}, \cdot) = \mathbf{Laz}(\mathfrak{a})$. Also $a \circ b = a \cdot \lambda_a(b)$ where $\lambda_a \in \text{Aut}(\mathfrak{a})$ is uniquely determined by $(a, \lambda_a) \in \gamma(\mathfrak{g})$. Since $\gamma(\mathfrak{a}) = \{(W(a), \exp(\mathcal{L}_a)) \mid a \in \mathfrak{a}\}$ we find $\lambda_a = \exp(\mathcal{L}_{\Omega(a)})$.

Also, $W : \mathbf{Laz}(\mathfrak{a}^\circ) \rightarrow (\mathfrak{a}, \circ)$ is a group isomorphism because it is the composition of the group homomorphisms $\mathbf{Laz}(\mathfrak{a}^\circ) \rightarrow \mathbf{Laz}(\mathfrak{g}) : a \mapsto (a, \mathcal{L}_a)$, $\gamma : \mathbf{Laz}(\mathfrak{g}) \rightarrow \gamma(\mathfrak{g})$ and $\text{pr}_{\mathfrak{a}} : \gamma(\mathfrak{g}) \rightarrow (\mathfrak{a}, \circ)$. $\square$

Remark 8.15. With the help of the explicit formula in Remark 5.17 find that

$$W(a) = a + \frac{1}{2}\mathcal{L}_a(a) + \frac{1}{6}\mathcal{L}_a^2(a) + \frac{1}{12}[a, \mathcal{L}_a(a)] + \frac{1}{24}(\mathcal{L}_a([a, \mathcal{L}_a(a)]) + \mathcal{L}_a^3(a)) + \dots$$

where further terms are of order at least 5.

We now explain how the above construction is related to the formal integration as described by Bai, Guo, Sheng and Tang in [5]. Let $\mathbb{F}$ be a field of characteristic 0. Then Bai, Guo, Sheng and Tang proved that to any connected complete post-Lie $\mathbb{F}$ -algebra $(\mathfrak{a}, \triangleright)$ one can associate a skew brace $(\mathfrak{a}, \cdot, \circ)$ where

$$a \cdot b = \text{BCH}(a, b), \quad a \circ b = a \cdot \exp(\mathcal{L}_{\Omega'(a)})(b).$$

We first note that in our terminology their notion of a connected complete post-Lie $\mathbb{F}$ -algebra is a connected complete post-Lie $\overline{\mathbb{F}}$ -algebra $(\mathfrak{a}, \triangleright)$ such that moreover $\mathfrak{a}_i \triangleright \mathfrak{a}_j \subseteq \mathfrak{a}_{i+j}$, or equivalently, such that the filtrations on $\mathfrak{a}$ and $\mathfrak{a}^\circ$ coincide. This is indeed a more restrictive notion, as for example if $\mathfrak{a}_i = 0$ for some i , then $\mathfrak{a}_{i-1}$ is contained in the socle hence $\text{Soc}(\mathfrak{a}, \triangleright) \neq 0$. In [17, Section 3] an example is given of a left-nilpotent pre-Lie $\mathbb{R}$ -algebra with trivial socle, so its associated pre-Lie $\overline{\mathbb{R}}$ -algebra, as in Example 6.3, is complete and connected in our sense but not in the sense of [5].

We quickly recall their construction: starting from a complete post-Lie $\overline{\mathbb{F}}$ -algebra $(\mathfrak{a}, \triangleright)$ such that $\mathfrak{a}_i^\circ = \mathfrak{a}_i$ for all $i \geq 0$ and consider the universal enveloping algebra $\mathcal{U}(\mathfrak{a})$. On $\mathcal{U}(\mathfrak{a})$ we define a new multiplication $*$ which for $a \in \mathfrak{a}$, $b \in \mathcal{U}(\mathfrak{a})$ is given by $a * b = ab + \delta_a(b)$, where δ_a is the unique derivation of $\mathcal{U}(\mathfrak{a})$ with $\delta_a(z) = a \triangleright z$ for $z \in \mathfrak{a}$. In this way, $(\mathcal{U}(\mathfrak{a}), *)$ now becomes an $\overline{\mathbb{F}}$ -algebra. This operation is then extended to the completion $\hat{\mathcal{U}}(\mathfrak{a})$ and $(\hat{\mathcal{U}}(\mathfrak{a}), *)$ filtration is thus a complete $\overline{\mathbb{F}}$ -algebra. For $a \in \mathfrak{a}$, we define $\Omega'(a) = \log_* \exp(a)$, where the exponential map with respect to the usual multiplication in $\hat{\mathcal{U}}(\mathfrak{a})$ is first applied, followed by the logarithmic map in $(\hat{\mathcal{U}}(\mathfrak{a}), *)$.

We prove that $W(a) = \log \exp_*(a)$, which implies that $\Omega = \Omega'$ and thus that our construction coincides with the one in [5]. Consider the two $\overline{\mathbb{F}}$ -algebra homomorphisms $s, t : \hat{\mathcal{U}}(\mathfrak{a}^\circ) \rightarrow \hat{\mathcal{U}}(\mathfrak{aff}(\mathfrak{a}))$ given by $s(a) = (a, \mathcal{L}_a)$ and $t(a) = (0, \mathcal{L}_a)$. We define

$$\phi : \hat{\mathcal{U}}(\mathfrak{a}^\circ) \rightarrow \hat{\mathcal{U}}(\mathfrak{aff}(\mathfrak{a})) : a \mapsto s(a_{(1)})t(S(a_{(2)})).$$

Then $\phi(g) \in \hat{\mathcal{U}}(\mathfrak{a}) \subseteq \hat{\mathcal{U}}(\mathfrak{aff}(\mathfrak{a}))$ for any grouplike element $g \in \mathbb{G}\hat{\mathcal{U}}(\mathfrak{a}^\circ)$, hence the image of ϕ is contained in $\mathcal{U}(\mathfrak{a}^\circ)$ since the grouplike elements span $\hat{\mathcal{U}}(\mathfrak{a}^\circ)$. Note that for $a \in \mathfrak{a}^\circ$, $b \in \hat{\mathcal{U}}(\mathfrak{a}^\circ)$,

$$\phi(ab) = \phi(a_{(1)})t(a_{(2)})(\phi(b)) = \phi(a) * \phi(b).$$

Hence

$$\exp_* \phi(a) = \phi \exp(a) = \exp(a, \mathcal{L}_a) \exp(0, \mathcal{L}_a)^{-1}$$

thus $\log \exp_*(a) = W(a)$.

In the pre-Lie case we have a more explicit description of W which coincides with the one in [2] and [36].

Proposition 8.16. *Let $\mathcal{R}$ be divisible and let $(\mathfrak{a}, \triangleright)$ be a complete connected pre-Lie $\mathcal{R}$ -algebra. Then $W(a) = \sum_{k=1}^{\infty} \frac{1}{k!} \mathcal{L}_a^{k-1}(a)$.*

Proof. We claim that under the above assumptions, if we consider $(\mathfrak{a}, \triangleright)$ as a complete connected post-Lie $\mathbb{Q}$ -algebra as in Example 6.3, W' coincides with the map W defined in Theorem 8.19 and thus that both constructions coincide in this case.

Define the $\mathcal{R}$ -module $M = R_0 \oplus \mathfrak{a}$ and consider the degree preserving Lie $\mathcal{R}$ -algebra homomorphism $\rho : \mathfrak{aff}(\mathfrak{a}) \rightarrow \text{End}_{\mathcal{R}}(M)$ as $\rho_{(a, \delta)}(r, b) = (0, ra + a \triangleright b)$. Similarly, define $\kappa : \text{Hol}(\mathbf{Laz}(\mathfrak{a})) \rightarrow \text{End}_{\mathcal{R}}(M)$ as $\kappa_{(a, \lambda)}(r, b) = (r, ra + \lambda(b))$ and note that this is a degree preserving homomorphism of filtered groups. Then, $\exp(\rho_{(a, 0)}) = \kappa_{(a, 0)}$ and $\exp(\rho_{(0, \delta)}) = \kappa_{(0, \exp(\delta))}$, hence $\gamma = \rho \exp \kappa^{-1}$. Since $\kappa_{(a, \lambda)}(1, 0) = a$, we find that

$$(0, W(a)) = \exp(\rho_{(a, \mathcal{L}_a)})(1, 0) = (0, \sum_{k=1}^{\infty} \frac{1}{k!} \mathcal{L}_a^{k-1}(a)). \quad \square$$

Proposition 8.17. *Let $\mathcal{R}$ be divisible and let $(A, \cdot, \circ)$ be a strict complete connected skew $\mathcal{R}$ -brace. Then*

$$\Omega : A \mapsto A : a \mapsto U(a, \lambda_a),$$

is a bijection and $\mathbf{L}(A, \cdot, \circ) = (\mathbf{Laz}^{-1}(A, \cdot), \triangleright)$ with $a \triangleright b = \log(\lambda_{W(a)})(b)$, is a complete connected post-Lie $\mathcal{R}$ -algebra, where $W := \Omega^{-1}$. Moreover, $\Omega : (A, \circ) \rightarrow \mathfrak{a}^\circ$ is a Lie $\mathcal{R}$ -algebra isomorphism, where $(\mathfrak{a}, \triangleright) = \mathbf{L}(A, \cdot)$.

Proof. Let $(A, \cdot, \circ)$ be a strict complete connected skew $\mathcal{R}$ -brace and let G be the associated regular sub $\mathcal{R}$ -group of $\text{Hol}(A, \cdot)^+$. Then Ω is a bijection since it is the composition of the bijections $A \mapsto G : a \mapsto (a, \lambda_a)$ and $U : G \rightarrow A$ with U as in Remark 8.7. The fact that U is bijective, follows from Proposition 5.20.

By Proposition 5.20 we find that $\gamma^{-1}(G)$ is a closed regular sub Lie $\mathcal{R}$ -algebra of $\mathfrak{aff}(\mathbf{Laz}^{-1}(A, \cdot))^+$, thus we obtain a complete connected post-Lie $\mathcal{R}$ -algebra

$(\mathbf{Laz}^{-1}(A, \cdot, \triangleright)$. We find $a \triangleright b = \mathcal{L}_a$ where $\mathcal{L}_a$ is uniquely determined by the fact that $(a, \mathcal{L}_a) \in \gamma^{-1}(G)$. Since $\gamma^{-1}(G) = \{(\Omega(a), \log(\lambda_a)) \mid a \in A\}$ we find that $\mathcal{L}_a = \log(\lambda_{\Omega^{-1}(a)})$.

Also, Ω is $(A, \circ) \rightarrow \mathbf{L}(A, \cdot, \circ)^\circ$ is an isomorphism of Lie $\mathcal{R}$ -algebras since it is the composition of the isomorphisms $(A, \circ) \rightarrow G : a \mapsto (a, \lambda_a)$, $\gamma^1 : G \rightarrow \gamma^{-1}(G)$ and $\text{pr}_A : \gamma^{-1}(G) \rightarrow \mathbf{Laz}^{-1}(A, \cdot)$. $\square$

Proposition 8.18. *Let $\mathcal{R}$ be divisible and let $(A, +, \circ)$ be a strict complete connected $\mathcal{R}$ -brace. Then $\Omega(a) = \sum_{k=1}^{\infty} \frac{1}{k} (-1)^{k+1} \lambda_a^{k-1}(a)$.*

Proof. Using the same notation as in the proof of Proposition 8.16, we find that $\rho_{(a, \delta)}(1, 0) = (0, a)$ hence

$$(0, \Omega(a)) = \log(\kappa_{(a, \lambda_a)})(1, 0) = (0, \sum_{k=1}^{\infty} \frac{1}{k} (-1)^{k+1} \lambda_a^{k-1}(a)). \quad \square$$

This formal differentiation actually coincides (under the necessary assumptions) with the geometric construction in [5, Theorem 4.3] (or in [10, Section 3]). Let $(\mathfrak{a}, \triangleright)$ be a finite dimensional complete connected post-Lie $\overline{\mathbb{F}}$ -algebra. Recall from [5, Theorem 4.3] that for a skew brace $(G, \cdot, \circ)$ such that $(G, \cdot)$ and $(G, \circ)$ are Lie groups and the map $a \mapsto (a, \lambda_a)$ is smooth (or in the original formulation, a post-Lie group) the authors define a post-Lie structure on $\mathfrak{g}$, the Lie algebra associated to $(G, \cdot)$, by starting with the map $\lambda : G \rightarrow \text{Aut}(G, \cdot) : a \mapsto \lambda_a$, which yields the map $\rho : G \rightarrow \text{Aut}(\mathfrak{g}) : a \mapsto d\lambda_a$. Subsequently, differentiating yields a map $d\rho : \mathfrak{g} \rightarrow \text{Der}(\mathfrak{g})$ and we define

$$a \triangleright' b = (d\rho)_a(b).$$

Let $(G, \cdot, \circ) = \mathbf{S}(\mathfrak{a}, \triangleright)$ for a finite dimensional complete connected post-Lie $\overline{\mathbb{R}}$ -algebra $(\mathfrak{a}, \triangleright)$, then $(G, \cdot)$ is an algebraic group so in particular it is a Lie group and $\mathfrak{a}$ is isomorphic to the Lie algebra associated to $(G, \cdot)$ such that under this identification, the exponential map $\mathfrak{a} \rightarrow G$ is just the identity. Thus, under this identification of $(G, \cdot)$ and $\mathfrak{a}$ we have $\lambda_g = \rho_g$. Hence, $b \mapsto a \triangleright' b$ is the differential of the map $\lambda : a \mapsto \exp(\mathcal{L}_{\Omega(a)})(b)$. We first determine $d\Omega$. Since $\Omega W = \text{id}$, it is sufficient to determine dW . By Remark 8.15, we find $W(a) = ta + \sum_{i=2}^{\infty} t^i a_i$ for some values $a_i \in \mathfrak{a}$, hence

$$dW(a) = \left. \frac{d}{dt} \right|_{t=0} ta + \sum_{i=2}^{\infty} t^i a_i = a,$$

from which we conclude that $d\Omega = \text{id}$. The map $\exp(\mathcal{L}) : \mathfrak{a} \rightarrow \text{Aut}(\mathfrak{a}) : a \mapsto \exp(\mathcal{L}_a)$ has differential

$$d(\exp(\mathcal{L}))(a) = \left. \frac{d}{dt} \right|_{t=0} \exp(\mathcal{L}_{ta}) = \mathcal{L}_a.$$

We conclude that $(d\lambda)_a = \mathcal{L}_a$ and thus $a \triangleright' b = a \triangleright b$, which proves the statement.

Theorem 8.19. *Let $\mathcal{R}$ be divisible, then*

$$\mathbf{S} : \mathcal{R}\mathbf{PLA}^{c,c} \rightarrow \mathcal{R}\mathbf{SB}^{c,c,s} \text{ and } \mathbf{L} : \mathcal{R}\mathbf{SB}^{c,c,s} \rightarrow \mathcal{R}\mathbf{PLA}^{c,c},$$

are mutually inverse functors. The map Ω associated to a complete connected post-Lie $\mathcal{R}$ -algebra $(\mathfrak{a}, \triangleright)$ coincides with the map Ω associated to $\mathbf{S}(\mathfrak{a}, \triangleright)$.

Closed sub post-Lie $\mathcal{R}$ -algebras of $(\mathfrak{a}, \triangleright)$ and closed strict sub skew $\mathcal{R}$ -braces of $\mathbf{S}(\mathfrak{a}, \triangleright)$ coincide and moreover, closed left ideals, closed strong left ideals and closed ideals of $(\mathfrak{a}, \triangleright)$ and $\mathbf{S}(\mathfrak{a}, \triangleright)$ coincide.

Proof. It is clear that $\mathbf{S}$ and $\mathbf{L}$ are mutually inverse constructions and that their associated maps Ω coincide. We prove the functoriality of $\mathbf{S}$, the proof for $\mathbf{L}$ is analogous.

Let $f : (\mathfrak{a}, \triangleright) \rightarrow (\mathfrak{b}, \triangleright)$ be a homomorphism of complete connected post-Lie $\mathcal{R}$ -algebras and set $(\mathfrak{a}, \cdot, \circ) = \mathbf{S}(\mathfrak{a}, \triangleright)$ and $(\mathfrak{b}, \cdot, \circ) = \mathbf{S}(\mathfrak{b}, \triangleright)$. Then by Theorem 5.21 $f : (\mathfrak{a}, \cdot) \rightarrow (\mathfrak{b}, \cdot)$ is an $\mathcal{R}$ -group homomorphism. As W is an isomorphism between $\mathbf{Laz}(\mathfrak{a}^\circ)$ and $(\mathfrak{a}, \circ)$, it is sufficient to proof $fW = Wf$ in order to conclude that $f : (\mathfrak{a}, \circ) \rightarrow (\mathfrak{b}, \circ)$ is a group homomorphism. Note that $f'(a, \mathcal{L}_a) = (f(a), \mathcal{L}_{f(a)})$ yields a homomorphism between the closed sub Lie $\mathcal{R}$ -algebras $\mathfrak{a} \oplus \{\mathcal{L}_a \mid a \in \mathfrak{a}\} \leq \mathfrak{aff}(\mathfrak{a})^+$ and $\mathfrak{b} \oplus \{\mathcal{L}_b \mid b \in \mathfrak{b}\} \leq \mathfrak{aff}(\mathfrak{b})^+$. In particular, we find that

$$\begin{aligned} fW(a) &= f \operatorname{pr}_a \operatorname{BCH}((a, \mathcal{L}_a)(0, -\mathcal{L}_a)) \\ &= \operatorname{pr}_b f' \operatorname{BCH}((a, \mathcal{L}_a)(0, -\mathcal{L}_a)) \\ &= \operatorname{pr}_b \operatorname{BCH}((f(a), \mathcal{L}_{f(a)})(0, -\mathcal{L}_{f(a)})) \\ &= Wf(a). \end{aligned}$$

The equivalence of closed sub post-Lie $\mathcal{R}$ -algebras and closed strict sub skew $\mathcal{R}$ -braces is a direct corollary of Theorem 8.19. In particular, note that for $\mathfrak{b}$ a closed sub post-Lie $\mathcal{R}$ -algebra of $(\mathfrak{a}, \triangleright)$ this implies that on the set $\mathfrak{b}$, the map $W_{\mathfrak{b}}$ belonging to $(\mathfrak{b}, \triangleright)$ coincides with the map W belonging to $(\mathfrak{a}, \triangleright)$, so in particular $W(\mathfrak{b}) = \mathfrak{b}$.

Clearly the closed sub Lie $\mathcal{R}$ -algebras of $\mathfrak{a}$ invariant under $\{\mathcal{L}_x \mid x \in \mathfrak{a}\}$ and $\{\exp(\mathcal{L}_x) \mid x \in \mathfrak{a}\}$ coincide, which yields the equivalence of closed left ideals. Taking into account Proposition 8.10, the equivalence of closed strong left ideals follows.

Let $\mathfrak{b}$ be a strong left ideal of $(\mathfrak{a}, \triangleright)$. As $W : \mathbf{Laz}(\mathfrak{a}^\circ) \rightarrow (A, \circ)$ is a group isomorphism, we find that $\mathfrak{b}$ is an ideal of $\mathfrak{a}^\circ$ if and only if $W(\mathfrak{b})$ is a normal sub $\mathcal{R}$ -group of $(A, \circ)$. If we recall that $W(\mathfrak{b}) = \mathfrak{b}$, the equivalence of ideals of $(\mathfrak{a}, \triangleright)$ and $\mathbf{S}(\mathfrak{a}, \triangleright)$ follows. $\square$

Remark 8.20. Recall that a skew brace $(A, +, \circ)$ is *square-free* if $a * a$ for all $a \in A$. Similarly, we say that a post-Lie ring $(\mathfrak{a}, \triangleright)$ is *square-free* if $a \triangleright a = 0$ for all $a \in \mathfrak{a}$. Note that the property of being square-free is preserved by the functors $\mathbf{S}$ and $\mathbf{L}$ and in particular, $W = \Omega = \operatorname{id}$ in this case. Therefore these classes behave particularly well and eliminate most of the computations involved.

Example 8.21. Let $\mathcal{R}$ be a divisible sequence and let A be a connected complete $\mathcal{R}$ -algebra. Then A is a pre-Lie algebra for $a \triangleright b = ab$ and it is well-known that the associated group of flows is given by $a \circ b = a + ab + b$. We can verify this directly: clearly $W(a) = \exp(a) - 1$, hence $\Omega(a) = \log(1 + a)$, where we for simplicity we work in the Dorroh extension $R_0 \oplus A$. We now find

$$a \circ b = a + \exp(\mathcal{L}_{\log(1+a)})(b) = a + ab + b.$$

Conversely, just working in $(A, +, \circ)$ we recover the original multiplication as

$$a \triangleright b = \log(\lambda_{W(a)})(b) = \log(\mathcal{L}_{1+W(a)})(b) = a + \log(\mathcal{L}_{\exp(a)})(b) = ab.$$

Recall that this is precisely the well-known correspondence between two-sided braces and Jacobson radical rings [30].

The following result extends [37, Proposition 15, Lemma 17].

Proposition 8.22. *Let $\mathcal{R}$ be divisible, let $(A, \cdot, \circ)$ be a complete connected skew $\mathcal{R}$ -brace and let n be an integer, then $\{a^n \mid a \in A\} = \{a^{\circ n} \mid a \in A\}$ and $\{a \in A \mid a^n = 1\} = \{a \in A \mid a^{\circ n} = 1\}$ and these sets are closed ideals of $(A, \cdot, \circ)$.*

Proof. We prove the statement for the first set, the second part is analogous. Let $(\mathfrak{a}, \triangleright)$ be the complete post-Lie $\mathcal{R}$ -algebra such that $(A, \cdot, \circ) = \mathbf{S}(\mathfrak{a}, \triangleright)$. Then $n\mathfrak{a}$ is a

closed ideal of $(\mathfrak{a}, \triangleright)$, hence it is also a closed ideal of $(A, \cdot, \circ)$ by Theorem 8.19. Now as $(A, \cdot) = \mathbf{Laz}(\mathfrak{a})$, we find $n\mathfrak{a} = \{a^n \mid a \in A\}$. Also, as $W : \mathbf{Laz}(\mathfrak{a}^\circ) \rightarrow (A, \circ)$ is an $\mathcal{R}$ -group isomorphism we find $W(n\mathfrak{a}) = \{a^{\circ n} \mid a \in A\}$, but as W maps any sub post-Lie $\mathcal{R}$ -algebra to itself, we conclude that $\{a^n \mid a \in A\} = \{a^{\circ n} \mid a \in A\}$. $\square$

Proposition 8.23. *Let $\mathcal{R}$ be divisible and let $(\mathfrak{a}, \triangleright)$ be a complete connected post-Lie $\mathcal{R}$ -algebra. Then $\text{Fix}(\mathfrak{a}, \triangleright) = \text{Fix}(\mathbf{S}(\mathfrak{a}, \triangleright))$, $\text{Soc}(\mathfrak{a}, \triangleright) = \text{Soc}(\mathbf{S}(\mathfrak{a}, \triangleright))$ and $\text{Ann}(\mathfrak{a}, \triangleright) = \text{Ann}(\mathbf{S}(\mathfrak{a}, \triangleright))$. In particular, if $\mathfrak{a}$ is nilpotent, then $(\mathfrak{a}, \triangleright)$ is right nilpotent if and only if $\mathbf{S}(\mathfrak{a}, \triangleright)$ is right nilpotent.*

Proof. The first part of the statement follows from the fact that $W(a) = a$ whenever $a \triangleright a = 0$, which in particular is the case when $a \in \text{Fix}(\mathfrak{a}, \triangleright)$ or $a \in \text{Soc}(\mathfrak{a}, \triangleright)$. The second part of the statement now follows from [1, Lemma 5.3 and Lemma 5.4] and the analogous statement for post-Lie algebras. $\square$

We are now ready to prove Theorem 1.5.

Proof of Theorem 1.5. Let $(\mathfrak{a}, \triangleright)$ be a post Lie p -ring of L -nilpotency class k . We can consider $(\mathfrak{a}, \triangleright)$ as a complete connected post-Lie $\overline{\mathbb{Z}}$ -algebra as in Example 6.3. Since $k < p$, it follows from a similar argument as used in the proof of Proposition 5.6 that this filtration also makes $(\mathfrak{a}, +)$ into a $\mathcal{Q}$ -module and the bilinear map $\mathfrak{a}^\circ \times \mathfrak{a} \rightarrow \mathfrak{a}$ is also a bilinear map of $\mathcal{Q}$ -modules. We conclude that $(\mathfrak{a}, \triangleright)$ becomes a complete post-Lie $\mathcal{Q}$ -algebra.

Conversely, let $(A, \cdot, \circ)$ be a skew p -brace of L -nilpotency class k . By Example 7.2 we find that the filtration $L^i(A)$ makes it into a complete connected skew $\overline{\mathbb{Z}}$ -brace. From Proposition 5.6 it follows that $(A, \cdot, \circ)$ is a strict complete connected skew $\mathcal{Q}$ -brace.

The first part of the statement now follows from Theorem 8.19 and the observation that the order of an element is preserved by the Lazard correspondence.

At last, recall that any skew brace of order p^k is left nilpotent by [12, Proposition 4.4] hence it is L -nilpotent. $\square$

9. DIFFERENTIATION IN NON-ZERO CHARACTERISTIC

Recall that a *primitive n -th root of unity* of a ring R is an element $\xi \in R$ such that $\xi^n = 1$ and $\xi^k - 1$ not a zero divisor for $1 \leq k < n$.

Lemma 9.1. *Let R be a ring, let M, N be R -modules, $\xi \in R$ a primitive n -th root of unity where n is invertible in R , and $f : M \rightarrow N$ a map such that for every $m \in M$ there exist elements $\alpha_0, \dots, \alpha_d \in N$, with $d < n$, such that*

$$f(rm) = \alpha_0 + r\alpha_1 + r^2\alpha_2 + \dots + r^d\alpha_d.$$

Then $\frac{1}{n} \sum_{i=0}^{n-1} \xi^{ik} f(\xi^{-i}m) = \alpha_k$ for all $1 \leq k \leq d$.

Proof. Evaluating the formula yields

$$\begin{aligned} \frac{1}{n} \sum_{i=0}^{n-1} \xi^{ik} f(\xi^{-i}m) &= \frac{1}{n} \sum_{i=0}^{n-1} \sum_{j=0}^d \xi^{i(k-j)} \alpha_j \\ &= f_k \end{aligned}$$

since $\sum_{i=0}^{n-1} \xi^{i(k-j)} f_j$ equals 0 for $j \neq k$ and it equals nf_k for $j = k$. $\square$

The above map should be seen as a discrete version of taking the k -th directional derivative through m and then evaluating at 0. Therefore, it seems justified to introduce the notation

$$df : M \rightarrow N : m \mapsto \frac{1}{n} \sum_{i=0}^{n-1} \xi^i f(\xi^{-i}m).$$

Although not stated explicitly, it is clear that this differentiation technique plays a prominent role in [34, 36]. We can extend the formula of [36, Theorem 12] to the skew brace setting.

Proposition 9.2. *Let p^k be a prime power, $\xi \in \mathbb{Z}/p^k\mathbb{Z}$ a primitive $p-1$ -th root of unity. Let $(A, \cdot, \circ)$ be a skew brace such that $(A, \cdot)$ has exponent p^k together and moreover $(A, \cdot)$ is a filtered group with finite filtration $A = A_0 = A_1 \supseteq \dots \supseteq A_p = 1$ and such that $A_i * A_j \subseteq A_{i+j}$ for all $i, j \geq 1$. Then*

$$a \triangleright b = \sum_{i=0}^{p-2} \xi^i \lambda_{\xi^{-i}a}(b),$$

for all $a, b \in A$ where the sum is taken in $\mathbf{Laz}^{-1}(A, \cdot)$.

Proof. Set $R = \mathbb{Z}/p^k\mathbb{Z}$. It is straightforward to verify that $(A, \cdot, \circ)$ with the natural filtration is a strict complete connected skew $\tilde{R}$ -brace, the strictness follows from the fact that every subgroup of an $\tilde{R}$ -group is a sub $\tilde{R}$ -group. By Theorem 8.19 we know that $(A, \cdot, \circ) = \mathbf{S}(\mathfrak{a}, \triangleright)$ for some complete connected post-Lie $\tilde{R}$ -algebra $(\mathfrak{a}, \triangleright)$. Consider the map $\lambda : \mathfrak{a} \rightarrow \text{End}_{\mathcal{R}}(\mathfrak{a}) : a \mapsto \lambda_a$. From Proposition 8.14 we find $\lambda_{ta} = \exp(\mathcal{L}_{W(ta)}) = \sum_{k=0}^{\infty} \sum_{i=1}^{\infty} \frac{1}{k!i!} \mathcal{L}_{\mathcal{L}_{ta}^{i-1}(ta)}^k$, but by the assumption on the filtration we find that all powers t^j disappear for $j \geq p$ and the degree 1 term is $t\mathcal{L}_a$, hence the assumptions for Lemma 9.1 are satisfied and we find $d\lambda(a) = \mathcal{L}_a$ which concludes the proof. $\square$

Note that a filtration as in Proposition 9.2 exists precisely if $A^{\{p\}} = 1$ where $A^{\{1\}} = A$ and $A^{\{k+1\}}$ is the subgroup of $(A, \cdot)$ generated by all $a * b$ and $aba^{-1}b^{-1}$ where $a \in A^{\{i\}}$, $b \in A^{\{k+1-i\}}$, $1 \leq i \leq k$. For braces this is precisely the chain of ideals $A^{[k]}$ used to define strong nilpotency as in [33].

We remark that there is one potential advantage of the above way of differentiating when working over characteristic p^k ; namely that we can have terms of degree more than p , whereas in the formulas for $\exp$ and $\log$ these terms are not allowed since the coefficients of degree p or more do not exist. Indeed, the field $\mathbb{F}_{p^n}$ for example has a primitive $(p^n - 1)$ -th root of unity. In particular, this means that there are possible uses in the case $p = 2$. It would be very interesting to see whether this idea of differentiation indeed has some meaningful application in such cases. In particular, the following question is natural to ask:

Question 9.3. Let $(A, +, \circ)$ be an $\mathbb{F}_{p^n}$ -brace such that $A^{[p^n-1]} = 1$ and let ξ be a primitive $(p^n - 1)$ -th root of unity in $\mathbb{F}_{p^n}$. Does the operation

$$a \triangleright b = \frac{1}{p^n - 1} \sum_{i=0}^{p^n-2} \xi^i \lambda_a(\xi^{-i}b),$$

yield a pre-Lie $\mathbb{F}_{p^n}$ -algebra on $(A, +)$?

ACKNOWLEDGEMENTS

The author was supported by Fonds Wetenschappelijk Onderzoek – Vlaanderen, via a PhD Fellowship fundamental research (1160524N) and a Senior Research Project (G004124N), and by Vrije Universiteit Brussel via the project OZR3762.

REFERENCES

- [1] E. Acri, R. Lutowski, and L. Vendramin. Retractability of solutions to the Yang–Baxter equation and p -nilpotency of skew braces. *Internat. J. Algebra Comput.*, 30(01):91–115, 2020.
- [2] A. A. Agrachev and R. V. Gamkrelidze. Chronological Algebras and Nonstationary Vector Fields. *J. Sov. Math.*, 17:1650–1675, 1981.

- [3] L. Auslander. Simply transitive groups of affine motions. *Amer. J. Math.*, 99(4):809–826, 1977.
- [4] D. Bachiller. Counterexample to a conjecture about braces. *Journal of Algebra*, 453:160–176, 2016.
- [5] C. Bai, L. Guo, Y. Sheng, and R. Tang. Post-groups, (Lie-)Butcher groups and the Yang–Baxter equation. *Mathematische Annalen*, 2023.
- [6] Y. Benoist. Une nilvariété non affine. *J. Differential Geom.*, 41(1):21–52, 1995.
- [7] F. Borceux. *Handbook of Categorical Algebra*. Encyclopedia of Mathematics and its Applications. Cambridge University Press, 1994.
- [8] D. Burde. Affine structures on nilmanifolds. *Internat. J. Math.*, 7(5):599–616, 1996.
- [9] D. Burde. Left-symmetric algebras, or pre-lie algebras in geometry and physics. *Open Mathematics*, 4(3):323–357, 2006.
- [10] D. Burde, K. Dekimpe, and S. Deschamps. Affine actions on nilpotent Lie groups. *Forum Mathematicum*, 21(5):921–934, 2009.
- [11] Cayley. On the Analytical Forms Called Trees. *Amer. J. Math.*, 4(1-4):266–268, 1881.
- [12] F. Cedó, A. Smoktunowicz, and L. Vendramin. Skew left braces of nilpotent type. *Proceedings of the London Mathematical Society*, 118(6):1367–1392, 2019.
- [13] L. N. Childs. On the Hopf Galois theory for separable field extensions. *Communications in Algebra*, 17(4):809–825, 1989.
- [14] S. Cicalò, W. A. De Graaf, and M. Vaughan-Lee. An effective version of the Lazard correspondence. *Journal of Algebra*, 352(1):430–450, 2012.
- [15] C. Curry, K. Ebrahimi-Fard, and H. Munthe-Kaas. What is a post-Lie algebra and why is it useful in geometric integration. In F. A. Radu, K. Kumar, I. Berre, J. M. Nordbotten, and I. S. Pop, editors, *Numerical Mathematics and Advanced Applications ENUMATH 2017*, pages 429–437, Cham, 2019. Springer International Publishing.
- [16] I. Del Corso. Module braces: relations between the additive and the multiplicative groups. *Annali di Matematica Pura ed Applicata*, 202(6):3005–3025, 2023.
- [17] D. Fried. Distality, completeness, and affine structures. *Journal of Differential Geometry*, 24(3):265–273, 1986.
- [18] L. Guarnieri and L. Vendramin. Skew braces and the Yang-Baxter equation. *Math. Comp.*, 86(307):2519–2534, 2017.
- [19] H. Kim. Complete left-invariant affine structures on nilpotent Lie groups. *Journal of Differential Geometry*, 24(3):373–394, 1986.
- [20] J.-L. Koszul. Domaines bornés homogènes et orbites de groupes de transformations affines. *Bull. Soc. Math. France*, 89:515–533, 1961.
- [21] M. Lazard. Sur les groupes nilpotents et les anneaux de Lie. *Annales scientifiques de l’École normale supérieure*, 71(2):101–190, 1954.
- [22] A. I. Malcev. Nilpotent torsion-free groups. *Izv. Akad. Nauk SSSR Ser. Mat.*, 13:201–212, 1949.
- [23] I. Martin-Lyons and P. J. Truman. Skew bracoids. *Journal of Algebra*, 638:751–787, 2024.
- [24] J. Milnor. On fundamental groups of complete affinely flat manifolds. *Advances in Math.*, 25(2):178–187, 1977.
- [25] M. F. Newman, E. A. O’Brien, and M. R. Vaughan-Lee. Groups and nilpotent Lie rings whose order is the sixth power of a prime. *Journal of Algebra*, 278(1):383–401, 2004.
- [26] D. Puljić. Classification of braces of cardinality p^4 . pages 1–18, 2022.
- [27] D. Puljić, A. Smoktunowicz, and K. Nejabati Zenouz. Some braces of cardinality p^4 and related Hopf-Galois extensions. *New York Journal of Mathematics*, 28:494–522, 2022.
- [28] D. Quillen. Rational Homotopy Theory. *The Annals of Mathematics*, 90(2):205, 1969.
- [29] W. Rump. Modules over braces. *Algebra Discrete Math.*, -(2):127–137, 2006.
- [30] W. Rump. Braces, radical rings, and the quantum Yang-Baxter equation. *J. Algebra*, 307(1):153–170, 2007.
- [31] W. Rump. The brace of a classical group. *Note di Matematica*, 34(1):115–144, 2014.
- [32] D. Segal. The structure of complete left-symmetric algebras. *Mathematische Annalen*, 293(1):569–578, 1992.
- [33] A. Smoktunowicz. On Engel groups, nilpotent groups, rings, braces and the Yang-Baxter equation. *Trans. Amer. Math.*, 370(9):6535–6564, 2018.
- [34] A. Smoktunowicz. A new formula for Lazard’s correspondence for finite braces and pre-Lie algebras. *Journal of Algebra*, 594:202–229, 2022.
- [35] A. Smoktunowicz. From pre-Lie rings back to braces. pages 1–24, 2022.
- [36] A. Smoktunowicz. On the passage from finite braces to pre-Lie rings. *Advances in Mathematics*, 409:108683, 2022.
- [37] A. Smoktunowicz. On the passage from finite braces to pre-Lie rings. *arXiv:2202.00085v3*, 2022.

- [38] L. Stefanello and S. Trappeniers. On the connection between Hopf–Galois structures and skew braces. *Bulletin of the London Mathematical Society*, 2023.
- [39] B. Vallette. Homology of generalized partition posets. *Journal of Pure and Applied Algebra*, 208(2):699–725, 2007.
- [40] E. B. Vinberg. The theory of homogeneous convex cones. *Trudy Moskov. Mat. Obšč.*, 12:303–358, 1963.
- [41] R. Warfield. *Nilpotent Groups*. Lecture Notes in Mathematics. Springer, 1976.

DEPARTMENT OF MATHEMATICS AND DATA SCIENCE, VRIJE UNIVERSITEIT BRUSSEL, PLEINLAAN 2, 1050 BRUSSELS, BELGIUM

Email address: `senne.trappeniers@vub.be`

URL: `https://sites.google.com/view/sennetrappeniers/`